Stated or Sampled? Post-Training Narrows a Language Model's Sampled Differential Diagnosis, Not the One It States

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Abstract

A language model's differential diagnosis can be read from the list it states in one answer or from the spread of its answers over repeated samples, the readout behind pass@k, self-consistency and sampling-based uncertainty. We compare the two readouts across base and post-trained versions of open 7–32B models from three families (Qwen, OLMo and Mistral) on DDXPlus, whose synthetic patients carry a reference differential, and on 897 published case reports. Post-training narrows the sampled differential in every family: post-trained models give 1.3–2.4 distinct answers per 32 samples, against 3.8–4.9, and with each model prompted at its best they lose 13–30 points of pass@32 (7–18 after subtracting what their samples reach against another patient's diagnosis). The stated list does not narrow. It stays close to the base model's averaged list for the same patient and varies little between calls, and for the smaller Qwen models it contains the true diagnosis more often. Severe conditions, also from published must-not-miss lists, are named more often on average, but specific ones are lost. Most of the first-answer gain of post-training disappears once base and post-trained models get the same prompt and decode greedily. On DDXPlus, majority voting stops helping and agreement between samples weakens as an error signal, so sampling-based tools understate what one post-trained answer contains; thinking mode restores agreement. On the case reports, post-trained samples likewise cover fewer of the alternatives the reports' authors considered while one stated list covers as many, but voting and agreement keep their value. Studies of diagnostic breadth should report which readout they use.

1 Introduction

Clinicians reason over a differential diagnosis, a ranked set of conditions consistent with the presentation, and the conditions they keep in view determine which tests are ordered and which dangerous possibilities are ruled out. Language-model evaluations in medicine nonetheless score a single answer: multiple-choice accuracy, top-1 diagnosis, or at most pass@k for a small k. Post-training, whether general instruction tuning and RLHF or domain-specific medical reinforcement learning and distillation, is reported to improve these scores [2, 12, 22].

A model's differential can be read in two ways: from the ranked list it states in one answer, or from the distribution of its answers over repeated samples. Much current practice uses the second. Pass@k, self-consistency voting [41] and sampling-based uncertainty such as semantic entropy [9] all draw on repeated samples, and so does the evidence that reinforcement learning sharpens models, which win at pass@1 while their base models solve more problems given many samples [30, 43, 46]. The two readouts agree for a model that samples from the distribution it states. Work outside medicine suggests that aligned models do not. They can verbalize distributions they fail to sample

from [27, 47], their verbalized confidence is better calibrated than their token probabilities [39], and their samples can homogenize enough to defeat sampling-based uncertainty [23]. Jang et al. [13] show this directly for survey-response distributions: instruction-tuned models sample worse than their own base models, more so at each post-training stage, while still describing the target accurately in a single call. We measure what this means for differential diagnosis, where a reference differential and a true diagnosis are available for every case. This is a replication and extension of that work in a clinical setting, not a new phenomenon; one aligned answer can also cover what many base-model samples spread out [19].

The design that holds the pretrained model fixed and varies only the post-training: (i) two base/instruct pairs, Qwen2.5-7B and Qwen3-8B, which isolate general post-training; (ii) three public medical models fine-tuned from the same parent, Qwen2.5-7B-Instruct: AlphaMed (rule-based RL), HuatuoGPT-o1 (SFT then PPO), and m1 (SFT distillation), plus AlphaMed's companion checkpoint trained from the base model; (iii) every released stage of OLMo 3 7B [38], from one base model through SFT, DPO and RL, plus a checkpoint trained with RL directly from the base; (iv) at 32B, Qwen2.5-32B with its Instruct model and every released stage of OLMo 3.1 32B. We draw 32 samples per case. On DDXPlus [8], which ships a reference differential for every synthetic patient, we score both readouts against that reference; on rare-disease cases from Phenopacket Store [6] we score accuracy and stated confidence.

Findings.

1. **Post-training collapses the sampled differential.** In both Qwen families at 7–8B the post-trained model is far better on its first answer (+13 to +27 points) but its pass@k curve crosses the base model's at k≈4–8, and at k=32 the base model leads by 10–15 points; at 32B the lead is smaller (4.0–4.9 points), though it grows to 15–19 points when every model sees three worked examples (§4.10). A third family, Mistral-7B prompted with worked examples, shows the same crossing, also with example lists from another model instead of the reference, whereas Qwen3-8B with its thinking mode on beats its base model at k=32 (§4.16, §4.17, §4.11). Part of the narrowing is sharper sampling: Qwen2.5-7B-Instruct sampled at T=1.3 matches its base model's pass@32 at T=0.7, while Qwen3-8B stays below it even at T=1.6. Most of the first-answer gain is prompt format and sampling noise: with the same worked examples the sampled pass@1 gain falls to 1.6–5.2 points in four of six pairs, and decoded greedily, post-training raises top-1 significantly only for Qwen3-8B and Mistral (§4.10, §4.12). Post-trained models give 1.3–2.6 distinct top answers per 32 samples, against 3.8–8.3, and can't-miss conditions drop out of their samples (§4.1, §4.2, §4.5). Which stage narrows the samples most depends on the recipe: the RL stage applied after SFT and DPO in OLMo 3 7B (19 points of pass@32), SFT in OLMo 3.1 32B (12 points), where the RL stage costs only 3 (§4.4).

2. **The stated list stays close to the base model's view and varies little between calls.** A single stated list from the post-trained Qwen models at 7–32B is closer to the reference differential than the base model's (total-variation distance −0.04 to −0.09), covers 6–11 points more of its mass, and names severe conditions 9–14 points more often on average, while their samples move away from the reference. The same holds for OLMo 3.1 32B. A post-trained list tracks its own base model's averaged list for the same patient and barely changes between calls, so post-trained lists stay close to the base model's view (for the stronger models; Table 21) and vary far less between calls; the post-trained Qwen lists contain the true pathology among their top five 12–24 points more often and allocate probability among

named conditions more like the reference, but five averaged base-model lists get about as close (§4.18). Severe conditions are named more often on average, but specific ones, most often acute pulmonary edema, are lost (§4.9). The gain survives restricting to complete lists, removing the true pathology, stratifying by top-1 correctness and correcting for each readout's floor (§4.9). OLMo 3 7B's five-item lists do not improve, but its stated distribution does when asked for a probability on every condition (§4.6, §4.8).

3. **Sampling-based methods lose their value; which signal to trust is model-dependent but cheap to check.** Majority voting over 32 samples gains base models 3–8 points at T=0.7 and the general post-trained models at most 1.9; models that reason before answering still gain 4–5. For post-trained Qwen models, the top three of one stated list beats the three most frequent of 32 sampled answers by 6–14 points. Agreement between samples becomes a weaker error signal in every family; the stated probability overtakes it for Qwen3-8B and OLMo-3-7B-Instruct but not for the other three general post-trained models. Averaging the two signals' ranks, with no labels, beats either signal for post-trained models (held-out AUROC 0.746 against 0.695 for agreement and 0.671 for the stated probability); a logistic model fitted on 100 labeled cases adds little (0.753). Qwen3-8B's thinking mode returns to the base-model pattern (§4.17). Neither readout tracks how ambiguous a case truly is (§4.7, §4.19).

4. **On real rare-disease cases the shifts replicate, though the top-three comparison does not flip.** Voting gains fall from 3.1–4.6 points for base models to 0.4–1.5 after post-training, and agreement weakens as an error signal until it ties the stated probability; 32 samples still beat one list (§4.23).

5. **Medical post-training adds accuracy, not breadth, and raises overconfidence on rare disease.** AlphaMed improves on its parent at every k without widening its sampled differential; its companion checkpoint trained from the base model keeps most of the base model's diversity (§4.20). With accuracy near 2%, medical RL raises stated confidence from 0.51 to 0.67–0.71, and every OLMo stage states 0.73–0.84 at about 1% accuracy (§4.21).

6. **RL from the base model need not collapse the samples.** GRPO from Qwen3-8B-Base nearly matches the official post-trained model's top-1 accuracy while keeping more than twice its sampled diversity, and a group-level reward narrows the samples less than standard RLVR at equal accuracy (§5).

2 Related work

What RL does to a model's output distribution. Yue et al. [46] find that RL with verifiable rewards improves pass@1 on math and code while base models win at large k. Wu et al. [43] argue that RLVR mostly reweights solutions already in the base model's support. Oh et al. [30] document a "sharpening tax" across 14 base/post-trained pairs on agentic benchmarks. In contrast, Cheng et al. [4] report that RL expands the boundary on domains under-represented in pretraining. In medicine, the only pass@k analysis we know of studies vision-language models and finds that RL mainly sharpens [14]. All of these measure the sampled distribution; we also measure the stated one.

Diversity loss from alignment. Instruction tuning and RLHF reduce output diversity [17, 28, 42], a loss that grows with scale for SFT [37]. Related work explains this as typicality bias in

preference data [47] and documents cross-model homogeneity [15]. Zhang et al. [47] also track diversity across the SFT, DPO and RLVR checkpoints of Tülu 3 on creative writing; we do the same for OLMo 3 against a clinical reference distribution.

Stated versus sampled distributions. Meister et al. [27] find that language models asked to verbalize a group's opinion distribution match it better than their samples or log-probabilities do, and that they "struggle to sample from their own distribution"; they study instruct and API models on opinion surveys, which have no correct answer. Verbalized sampling [47] prompts aligned models to state a distribution over candidate responses and recovers part of the diversity lost in post-training. Tian et al. [39] show that RLHF models' verbalized confidence is better calibrated than their token probabilities. Liu [23] report that aligned models' samples often fall into a single semantic cluster, leaving sampling-based uncertainty [9, 18] without signal on those questions; in a base/SFT/DPO stage ablation the collapse appears at DPO. Closest to our question, Jang et al. [13] show on synthetic and survey targets that instruction-tuned models sample worse than their own base models in every family they compare, that the gap grows at each OLMo-2 post-training stage, and that the same models describe the target accurately in one call. Verbalized-confidence studies reach mixed conclusions: sample consistency usually discriminates better than verbalized confidence [44], including for medical diagnosis [35]. Our stated readout is the verbalized top-k elicitation of Tian et al. [39], and hybrids of consistency and verbalized confidence usually beat either [24, 33, 44]. We add a clinical task with a reference differential and a true diagnosis, two pipelines at 7–32B, and the consequences for aggregation and uncertainty.

Medical reasoning models and calibration. Medical post-training recipes include verifier-guided SFT and PPO [2], minimalist rule-based RL [22], and test-time-scaling distillation [12]. Reward models and RL tend to favor confident answers [5, 20], and reasoning models are often overconfident [26]. LLMs have been evaluated as differential-diagnosis aids on clinicopathological cases [16, 25] and in conversation [40], and base models have been excluded from clinical evaluations for failing to follow instructions [11], the format problem our worked examples address. We show that this overconfidence appears in public medical descendants of one parent, on rare disease.

Differential diagnosis benchmarks and distributional RL. DDXPlus [8] was introduced for automatic diagnosis systems trained on differentials [7]. Recent LLM evaluations on it score hierarchical top-k accuracy [21] or interactive diagnosis [34]; the dataset's own metrics compare predicted and reference differentials as sets. Puri et al. [32] train models with GRPO to output several answers per forward pass, rewarding coverage of a gold *set* and set-level calibration, and evaluate on DDXPlus. We instead compare stated and sampled differentials against the reference *distribution* across public post-trained model families. Our group-level reward operates on the distribution of independently sampled answers (§5).

3 Experimental setup

Tasks. **DDXPlus** [8]: synthetic patients produced by a rule-based medical knowledge base. Each patient has a true pathology and a reference differential, a probability distribution over the 49 conditions. We use 980 test patients, 20 per condition. Each patient's coded symptoms and history are rendered as a short vignette; we fix known mistranslations in the English evidence descriptions. The prompt lists all 49 conditions and asks for the five most likely, with probabilities, as JSON. **Phenopacket Store** [6] (v0.1.27): published rare-disease cases annotated with HPO phenotypes

Model	Parent	Post-training	Role
Qwen2.5-7B	n/a	none (pretrained)	base
Qwen2.5-7B-Instruct	Qwen2.5-7B	general SFT + RLHF	instruct; medical parent
AlphaMed-7B [22]	Qwen2.5-7B-Instruct	rule-based RL	medical
HuatuoGPT-o1-7B [2]	Qwen2.5-7B-Instruct	SFT + PPO (verifier)	medical
m1-7B-23K [12]	Qwen2.5-7B-Instruct	SFT distillation	medical
Qwen3-8B-Base	n/a	none (pretrained)	base
Qwen3-8B	Qwen3-8B-Base	general post-training	instruct
Qwen2.5-32B	n/a	none (pretrained)	base
Qwen2.5-32B-Instruct	Qwen2.5-32B	general SFT + RLHF	instruct
OLMo-3-1025-7B [38]	n/a	none (pretrained)	base
OLMo-3-7B-Instruct-SFT	OLMo-3-1025-7B	SFT	stage 1
OLMo-3-7B-Instruct-DPO	OLMo-3 SFT	DPO	stage 2
OLMo-3-7B-Instruct	OLMo-3 DPO	RLVR	stage 3 (instruct)
OLMo-3-7B-RL-Zero-General	OLMo-3-1025-7B	RL (LM judge)	RL from base
OLMo-3-1125-32B	n/a	none (pretrained)	base
OLMo-3.1-32B-Instruct-SFT	OLMo-3-1125-32B	SFT	stage 1
OLMo-3.1-32B-Instruct-DPO	OLMo-3.1 SFT	DPO	stage 2
OLMo-3.1-32B-Instruct	OLMo-3.1 DPO	RLVR	stage 3 (instruct)

Table 1: Models studied. Each pair or family shares its pretrained weights, so differences reflect post-training. OLMo 3 and OLMo 3.1 release every stage of their instruct pipelines.

and an OMIM diagnosis. We use a disease-balanced set of 1,401 cases covering 749 diseases (at most two per disease). Each case is a vignette of present and absent phenotypes, and the model is asked for an open-ended differential. Answers are graded by mapping names to Mondo; we credit an exact match, and separately a "related" match to a specific ancestor group or subtype.

Models. Table 1. All checkpoints are public. For the medical models we verified the parent model from each released configuration.

Sampling. We draw n=32 samples per case with vLLM, at T=0.7, top-p 0.95, and at T=1.0, top-p 1.0 (sampling from the model's own distribution). All models use their chat template and the same prompt; OLMo-3-1025-7B ships without one, so it gets its SFT checkpoint's template. Completions are capped at 4,096 tokens (8,192 for m1, whose long reasoning otherwise truncates), and generation stops at the end-of-turn token. The 32B models are sampled at T=0.7 only; the OLMo 3.1 32B SFT and DPO checkpoints, stored in float32, are run in bfloat16 like the other stages. We take the last valid JSON list in each completion. Unparsable completions count as incorrect; this affects mainly base models, so base-model pass@k is a lower bound at large k, while post-trained gains at small k are partly format. §4.3 reports format-controlled comparisons.

Metrics. *pass@k*: the unbiased estimator of Chen et al. [3] on the top-1 answer of each sample. *Distinct* and *entropy*: the number of distinct top-1 diagnoses across a case's 32 samples, and the entropy of their empirical distribution, averaged over cases. On DDXPlus, TV: total-variation distance between this empirical distribution and the reference differential. *Coverage@k*: the expected share of the reference differential's probability mass covered by the distinct top-1 answers in k random samples. *Calibration*: expected calibration error (10 bins) of the stated top-1 probability

Pair	Model	pass@1	pass@32	Distinct	Entropy	TV↓	Cov@32	ECE↓
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**1.31**	**0.712**	**0.407**	0.214
	post-trained	**0.556**	0.643	1.53	0.18	0.783	0.236	**0.061**
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**1.58**	**0.715**	**0.422**	0.303
	instruct	**0.464**	0.663	2.65	0.49	0.761	0.290	**0.137**
Qwen3-8B, T=0.7	base	0.388	**0.730**	**3.76**	**0.78**	**0.750**	**0.322**	0.121
	post-trained	**0.558**	0.607	1.27	0.11	0.793	0.217	**0.063**

Table 2: DDXPlus, 32 samples per case (980 cases). The reference differential has mean entropy 1.87. Distinct and entropy refer to the top-1 answers across samples; Cov@32 is the share of reference probability mass covered by those answers. ECE is for the stated top-1 probability.

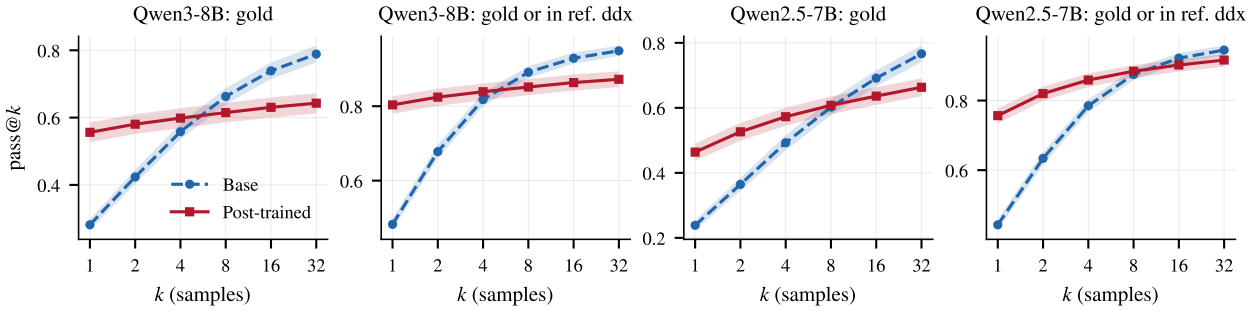


Figure 1: pass@k on DDXPlus at T=1 (980 cases, 32 samples), base model (dashed) vs. its post-trained version (solid), with 95% case-bootstrap bands. "Gold": the top-1 answer is the true pathology; "gold or in ref. ddx": it is the true pathology or appears in the reference differential. In both families the post-trained model wins at small k and loses at large k.

against top-1 correctness. These metrics describe the *sampled* readout. §4.6 defines the corresponding metrics for the *stated* readout, the list in a single completion. Differences come with 95% paired-bootstrap intervals over cases (2,000 resamples) and are computed from unrounded values. We report many intervals without multiplicity correction; we treat a non-significant difference as "no detectable change", with the interval bounding its size, not as equivalence.

4 Results

4.1 General post-training trades coverage for top-1 accuracy

Table 2 and Figure 1 compare each base model with its post-trained counterpart on DDXPlus. In both families the post-trained model is much better on its first answer: +27.5 points for Qwen3 and +22.5 for Qwen2.5 at T=1 (intervals $[+25.3, +29.6]$ and $[+20.6, +24.5]$). The advantage shrinks with k and reverses: the curves cross at $k \approx 4$–8, and at k=32 the base model leads by 14.6 points (Qwen3, $[+12.2, +17.1]$) and 10.3 points (Qwen2.5, $[+8.0, +12.8]$). The pattern is the same at T=0.7 for Qwen3, crossing at k=4 with a base lead of 12.4 points at k=32 ($[+10.1, +14.9]$). Unparsable base-model samples count as wrong, so these gaps are lower bounds. Qwen2.5 at T=0.7 behaves the same way: +12.6 points at k=1 $[+10.7, +14.5]$ and a base lead of 9.7 at k=32 $[+7.5, +11.9]$.

4.2 The sampled differential collapses, while top-1 calibration improves

The gap at large k reflects a collapsed *sampled* differential. Across 32 samples, the post-trained Qwen models name 1.3–2.7 distinct diagnoses per case, against 3.8–8.2 for their base models. Their answer entropy is 0.11–0.49, against 0.78–1.58 for base models and 1.87 for the reference differential (Table 2). Sampling at T=1 widens the base models' answers but barely changes the post-trained models' (Qwen3 entropy 0.11 at T=0.7, 0.18 at T=1), so the collapse lives in the model's own distribution rather than in the decoding settings. As a result, post-trained models cover less of the reference differential: 0.24 vs. 0.41 of its probability mass at k=32 for Qwen3, and 0.29 vs. 0.42 for Qwen2.5. Their top-1 calibration, however, is markedly better (ECE 0.06 vs. 0.21 and 0.14 vs. 0.30). This concerns verbalized confidence; alignment is known to worsen the calibration of token probabilities [31, 48], while verbalized confidence of RLHF models is better calibrated [39]. On every single-answer criterion (accuracy, small-k pass@k, calibration), the post-trained model therefore looks strictly better, and the loss of breadth in its samples is invisible. Whether the breadth is lost from the model's stated differential as well is a separate question (§4.6).

Which diagnoses drop out? For each case and each condition that the reference differential weights at least 0.1, we ask whether the model names that condition as its top answer in at least one of its 32 samples. We call this *considering* the condition. We group the 49 conditions by DDXPlus severity (1 = most severe). Post-training lowers consideration in every tier (Table 3), and the severe tier (DDXPlus severity 1–2, the dataset's label rather than a clinical consensus list) is not spared. This metric needs a chance reference, though: a guesser that picks uniformly among the 49 conditions scores about 0.48 on it, and one that draws from the average reference differential 0.56–0.61, above every model, base models included. Considered@32 therefore measures the breadth of the sampled answers, not clinical judgement; we assess severe conditions on the stated list instead (§4.9). For the 17 severity-1–2 conditions (e.g. NSTEMI/STEMI, acute pulmonary edema, anaphylaxis, pulmonary embolism, unstable angina), consideration falls from 0.45 to 0.27 for Qwen3 and from 0.51 to 0.34 for Qwen2.5 at T=1. The differences are -18.4 $[-20.8, -15.9]$ and -17.0 $[-19.7, -14.2]$ points. Some diagnoses disappear from the top answer almost entirely. Qwen2.5-7B-Instruct names unstable angina as its top answer in one of 107 eligible cases, where its base model does in 47%. When unstable angina is the true pathology, pass@32 falls from 1.00 to 0.00. Stable angina (0.89 to 0.06) and myocarditis (0.41 to 0.00) follow the same pattern. These conditions are not forgotten: they still appear lower in the listed differential (unstable angina in 50% of eligible cases, against 95% for the base model). The model ranks them but no longer commits to them. At the same time, post-trained models concentrate on a few salient "attractor" diagnoses. Qwen2.5-7B-Instruct gives NSTEMI/STEMI as its top answer in 13.1% of samples, against a reference rate of 4.4% and 6.4% for its base model. It also over-predicts pulmonary embolism (8.2% vs. 3.2%) and pneumonia (8.0% vs. 2.4%). Chest-pain presentations thus collapse onto the most dangerous candidate. That raises its accuracy (NSTEMI pass@1 0.99 vs. 0.44) at the cost of the alternatives. Ruling out acute coronary syndrome first is standard practice for chest pain, but these vignettes lack the ECG and biomarkers that would separate the coronary conditions, so we read this as a concentration of the samples, not as a clinical error.

4.3 Robustness to prompt and output format

Prompt. We repeat the Qwen3-8B pair with two other prompts (Table 4). *Plain* drops the system prompt and rewords the instructions. *CoT* asks the model to reason step by step before answering; Qwen3's thinking mode stays off in all runs. The plain prompt reproduces the pattern:

Comparison	Considered@32 (base → post-trained) Severe (1–2)	Moderate (3)	Mild (4–5)
Qwen3-8B, T=1	0.45 → 0.27	0.50 → 0.32	0.45 → 0.20
	−18.4 [−20.8, −15.9]	−18.1 [−20.7, −15.7]	−24.2 [−27.0, −21.4]
Qwen2.5-7B, T=1	0.51 → 0.34	0.50 → 0.37	0.38 → 0.23
	−17.0 [−19.7, −14.2]	−12.1 [−14.7, −9.7]	−15.0 [−17.4, −12.6]
Qwen3-8B, T=0.7	0.34 → 0.24	0.44 → 0.29	0.31 → 0.18
	−9.8 [−11.8, −7.9]	−15.8 [−18.2, −13.5]	−13.4 [−15.8, −11.1]
Qwen2.5-7B, T=0.7	0.40 → 0.28	0.41 → 0.31	0.28 → 0.17
	−12.4 [−14.7, −10.2]	−9.1 [−11.2, −6.7]	−10.3 [−12.3, −8.3]
Instruct → AlphaMed, T=0.7	0.28 → 0.32	0.31 → 0.34	0.17 → 0.21
	+4.3 [+2.0, +6.6]	+2.2 [+0.4, +4.0]	+3.9 [+2.4, +5.4]
Instruct → HuatuoGPT-o1, T=0.7	0.28 → 0.36	0.31 → 0.37	0.17 → 0.27
	+8.4 [+6.1, +10.8]	+6.0 [+4.0, +7.9]	+9.5 [+7.5, +11.5]
Instruct → m1, T=0.7	0.28 → 0.41	0.31 → 0.41	0.17 → 0.29
	+13.5 [+10.8, +16.2]	+9.7 [+7.2, +12.2]	+11.4 [+9.2, +13.6]
Base → AlphaMed (base-RL), T=0.7	0.40 → 0.39	0.41 → 0.40	0.28 → 0.27
	−0.7 [−3.0, +1.7]	−0.4 [−2.6, +1.7]	−0.4 [−2.4, +1.7]
Base → `acc` GRPO (8B), T=0.7	0.34 → 0.31	0.44 → 0.42	0.31 → 0.27
	−2.8 [−4.4, −1.3]	−2.1 [−3.7, −0.6]	−4.3 [−6.2, −2.5]
Base → `gm` GRPO (8B), T=0.7	0.34 → 0.32	0.44 → 0.43	0.31 → 0.28
	−1.5 [−3.2, +0.1]	−1.3 [−3.0, +0.3]	−3.0 [−4.8, −1.2]

Table 3: Share of (case, condition) pairs with reference probability ≥ 0.1 in which the condition is a top-1 answer in at least one of 32 samples, by DDXPlus severity. Second rows give differences in points with 95% case-bootstrap intervals. All rows use the 980 test cases; GRPO rows are seed 0. General post-training lowers consideration in every tier, including the severity-1–2 tier; medical post-training raises it; our GRPO from the base model loses at most 4.3 points, and with the group reward the severe-tier loss is not significant.

the post-trained model gains +23.0 [+20.9, +25.1] points at k=1 and loses −7.1 [−9.3, −5.0] at k=32 (T=0.7; at T=1, +34.2 and −8.7 [−11.0, −6.4]), with 1.3–1.5 distinct answers against 3.7–5.6. Asking for reasoning changes the picture. With CoT, the post-trained model's pass@32 rises from 0.607 to 0.835: it leads the base model at every k at T=0.7 (+4.7 [+2.9, +6.6] at k=32) and ties it at T=1 (0.0 [−2.1, +1.9]). Its samples remain far less diverse than the base model's (2.1–2.2 vs. 3.9–6.3 distinct answers), so reasoning recovers coverage at k=32 without restoring sampling diversity. The coverage loss is therefore largest when the model answers directly, and reasoning at inference time mitigates it. This matches the medical models: the reasoning-trained descendants widen the differential (§4.20).

Output format. Base models more often give invalid answers: unparsable, or with a top answer outside the 49 conditions. This affects 12% of Qwen3-8B-Base samples at T=0.7 and 25% at T=1 (Qwen2.5-7B: 1% and 21%), against at most 0.3% for the post-trained models. Invalid samples count as wrong, which deflates the base model at large k but inflates the post-trained model's gain at small k. Among valid samples only, the post-trained models still gain +12.0 [+9.8, +14.1] (Qwen3, T=0.7), +18.7 [+16.6, +20.7] (Qwen3, T=1), +12.2 [+10.4, +14.0] and +16.8 [+15.0, +18.5] points (Qwen2.5) at k=1, so a quarter to a third of the raw gain at T=1 is format. On the cases where every sample of both models is valid, the large-k loss persists: −15.2 [−21.9, −8.6] at k=32 for

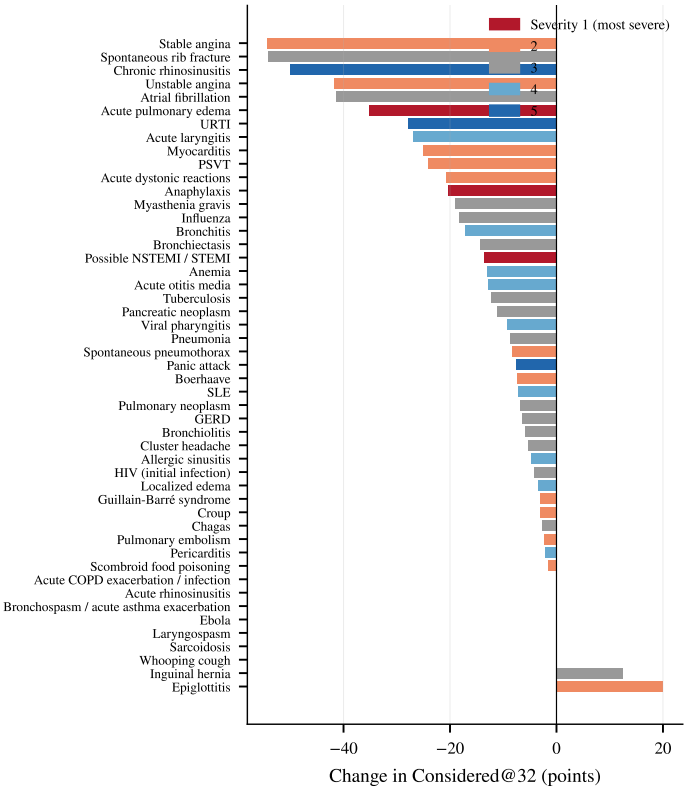


Figure 2: Change in Considered@32 per condition, Qwen2.5-7B-Instruct minus its base model (T=1, 980 cases), colored by DDXPlus severity (dark red = most severe). Conditions with fewer than five cases giving them reference probability ≥ 0.1 are omitted. Angina, acute pulmonary edema, anaphylaxis and myocarditis are among the largest drops.

Qwen3 (105 cases) and -10.2 $[-12.7, -7.6]$ for Qwen2.5 (748 cases), both at T=0.7. (At T=1 almost no case has 32 valid base samples.) The reported large-k losses are thus lower bounds.

4.4 Which post-training stage narrows the differential?

The Qwen pairs compare a base model with the end of a post-training pipeline. OLMo 3 [38] releases every stage of its 7B instruct pipeline from one base model: supervised fine-tuning, then DPO, then RL with verifiable rewards on math, code, instruction-following and chat prompts (the released OLMo-3-7B-Instruct). It also releases RL-Zero checkpoints trained with RL directly from the base; we use the general one, trained on chat prompts with an LM-judge reward. (The RL-Zero-Mix checkpoint does not load in our vLLM version.) Table 5 gives the results.

The coverage loss replicates in OLMo. At T=0.7, every post-trained stage is worse than the base at k=32: by -14.6 $[-17.2, -11.6]$ points after SFT, -9.2 $[-12.1, -6.1]$ after DPO, -28.3 $[-31.4, -25.1]$ after RLVR, and -13.3 $[-16.1, -10.5]$ for RL-Zero. Diversity falls at every stage of the pipeline, from 4.93 distinct answers for the base to 4.01 after SFT, 3.30 after DPO and 2.37 after RLVR.

Compared stage to stage, DPO recovers part of what SFT lost at large k ($+5.4$ $[+2.9, +8.1]$ points at k=32, and $+10.3$ $[+8.9, +11.6]$ at k=1) while narrowing the differential further (-0.71 distinct answers). The final RLVR stage is the largest single narrowing. Against the DPO model it loses -19.1 $[-21.9, -16.5]$ points at k=32, 0.92 distinct answers and 7.7 points of coverage@32,

		pass@1		pass@32		Distinct	
Prompt	T	Base	Post	Base	Post	Base	Post
default	0.7	0.388	**0.558**	**0.730**	0.607	3.76	1.27
default	1.0	0.282	**0.556**	**0.789**	0.643	6.60	1.53
plain	0.7	0.352	**0.583**	**0.727**	0.655	3.72	1.28
plain	1.0	0.244	**0.586**	**0.792**	0.705	5.57	1.54
CoT	0.7	0.409	**0.652**	0.788	**0.835**	3.89	2.08
CoT	1.0	0.275	**0.642**	0.837	0.837	6.31	2.24

Table 4: Qwen3-8B-Base vs. Qwen3-8B on DDXPlus (980 cases, 32 samples) under three prompts. Bold marks the significantly better model (paired case bootstrap). Without reasoning, the base model wins at k=32; with CoT, the post-trained model matches or beats it, but its samples stay far less diverse.

and it is also worse at k=1 (-5.9 $[-7.1, -4.7]$). It drops severe-tier Considered@32 to 0.25, a loss of -16.1 $[-18.6, -13.5]$ points relative to the base, against -7.4 for the DPO model. At T=1 the pattern is the same (RLVR vs. DPO: -12.7 $[-15.4, -9.9]$ at k=32).

RL applied directly to the base narrows the differential less. At T=1, RL-Zero shows no detectable loss of the base model's pass@32 (-1.8 $[-4.4, +0.7]$; clustered $[-5.8, +2.1]$) and 6.28 of its 7.53 distinct answers, and it loses 4.0 $[1.2, 6.9]$ points of severe-tier consideration, against 14.4 for the full instruct pipeline. At T=0.7 it still loses 13.3 points at k=32 (clustered $[7.3, 19.7]$), about as much as SFT.

The OLMo base model often breaks the output format: 40% of its samples are invalid at T=0.7 and 47% at T=1, while unparsable completions are 16–21% for the SFT model and at most 1.6% for the later stages. No case has 32 valid base samples, so the all-valid check of §4.3 is not possible here. Among valid samples, the post-trained models' gains at k=1 are small (-1.9 to $+4.0$ points at T=0.7, $+1.3$ to $+8.1$ at T=1), so most of OLMo's raw gain at k=1 is format. Invalid samples count as misses and are excluded from distinct answers and entropy, so the base model's coverage and diversity are understated and the losses above are conservative.

Across these pipelines, then, the collapse of the sampled differential is not a necessary effect of RL. It accumulates over the instruction-tuning pipeline; in OLMo 3 7B its largest step is RL applied on top of SFT and DPO, but at 32B it is SFT (§4.5), so which stage narrows most depends on the recipe. RL applied directly to a base model (OLMo RL-Zero, AlphaMed's base-initialized checkpoint, and our GRPO runs in §5) narrows the differential less. On Phenopacket Store the instruct pipeline also loses coverage: every stage trails the base model at k=32 (-2.0 $[-3.0, -1.1]$ points after SFT, -1.5 after DPO, -1.4 after RLVR; all significant), while RL-Zero does not (-0.6 $[-1.4, +0.2]$), from a base model at 4.5% (§4.21).

4.5 Replication at 32B

We repeat the DDXPlus comparisons at T=0.7 with Qwen2.5-32B and its Instruct model, and with every released stage of OLMo 3.1 32B, whose base is OLMo-3-1125-32B; there is no 32B RL-Zero checkpoint (Table 6). The sampled differential collapses at 32B too, by about half as much. Post-training costs 4.9 $[2.9, 7.1]$ points at pass@32 for Qwen2.5-32B and 4.0 $[2.0, 6.0]$ for OLMo 3.1 32B (clustered $[-0.5, 10.3]$ and $[1.2, 6.8]$; neither survives correction), halves the number of distinct answers (3.78 $\rightarrow$ 1.68 and 4.32 $\rightarrow$ 2.15), and lowers severe-tier consideration by 6.1 and 8.4 points. For Qwen the loss holds on the 421 cases where every base sample is valid (-4.8 $[-7.8, -1.7]$); the OLMo 32B base gives invalid answers on 27% of samples. With few-shot prompts, which nearly

T	Stage	pass@1	pass@32	Distinct	Entropy	Cov@32	Severe C@32
0.7	Base	0.178	**0.705**	**4.93**	**1.16**	**0.351**	**0.412**
	SFT	0.222	0.559	4.01	0.87	0.257	0.373
	SFT+DPO	**0.324**	0.613	3.30	0.63	0.272	0.338
	SFT+DPO+RLVR (Instruct)	0.265	0.422	2.37	0.42	0.195	0.251
	RL-Zero (RL from base)	0.295	0.572	4.06	0.84	0.286	0.326
1.0	Base	0.128	**0.713**	**7.53**	**1.66**	**0.398**	**0.469**
	SFT	0.191	0.634	6.32	1.29	0.321	0.426
	SFT+DPO	**0.313**	0.688	4.48	0.87	0.312	0.410
	SFT+DPO+RLVR (Instruct)	0.272	0.561	3.58	0.68	0.259	0.325
	RL-Zero (RL from base)	0.279	0.695	6.28	1.21	0.367	0.429

Table 5: OLMo 3 7B post-training stages on DDXPlus (980 cases, 32 samples). Severe C@32 is severe-tier Considered@32 as in Table 3. Bold marks the best value in each block. Diversity falls at every stage of the instruct pipeline, and the RLVR stage loses the most coverage; RL from the base keeps the base model's pass@32 at T=1.

Model	pass@1	pass@32	Distinct	Cov@32	Severe C@32	TV one list↓	TV samples↓	Vote@32 gain
Qwen2.5-32B (base)	0.500	**0.795**	**3.78**	**0.352**	**0.35**	0.726	**0.726**	+7.0
Qwen2.5-32B-Instruct	**0.604**	0.746	1.68	0.276	0.29	**0.674**	0.759	+1.9
OLMo-3-1125-32B (base)	0.326	**0.796**	**4.32**	**0.347**	**0.39**	0.746	**0.742**	+8.4
SFT	0.461	0.674	2.94	0.265	0.27	0.736	0.776	+3.6
SFT+DPO	**0.584**	0.784	2.29	0.293	0.32	**0.694**	0.755	+2.1
SFT+DPO+RLVR (Instruct)	0.560	0.756	2.15	0.283	0.30	0.701	0.763	+0.9

Table 6: 32B replication on DDXPlus (T=0.7, 980 cases, 32 samples). TV columns: distance to the reference differential of one stated list and of the sampled top-1 answers (§4.6). Vote@32 gain: accuracy of the majority answer over 32 samples minus single-call accuracy, in points (§4.7). Bold: best value per family.

remove invalid answers, both losses are three to five times larger (§4.10). The stage attribution differs from 7B. In OLMo 3.1 32B the largest narrowing is SFT (pass@32 −12.1 [−14.5, −9.8], clustered [−18.0, −6.8], distinct answers −1.39, severe tier −11.9); DPO recovers most of it (+10.9 against SFT), and the final RL step is small (−2.8 [−4.4, −1.2] at k=32, clustered [−5.3, −0.5], −0.14 distinct answers). OLMo 3.1 32B and OLMo 3 7B were post-trained with different recipes, so the stage that narrows the samples most is a property of the pipeline, not of SFT, DPO or RL as such. Under this prompt the stated readout improves in both 32B families (§4.6; with few-shot prompts only Qwen2.5-32B's does robustly, §4.10), and voting stops paying off: majority voting over 32 samples gains the base models 7.0 and 8.4 points and the Instruct models 1.9 and 0.9 (the latter n.s.).

4.6 The stated differential does not collapse

Every completion also *states* a differential: up to five conditions with probabilities. We read it directly. For each valid sample we renormalize its stated probabilities over the conditions it names (uniform if the probabilities are missing or invalid) and compare this single list with the reference differential: its total-variation distance to the reference, the reference mass of the conditions it names, and, for each severe condition with reference probability at least 0.1, whether the list names it. For the sampled readout we use the empirical distribution of top-1 answers over the valid

samples and, for the same budget of five names, the reference mass covered by the distinct top answers of five random samples. Two oracles bound what each format can achieve: the reference's own top five stated as a list, and as many draws from the reference as the model has valid samples (Table 7, Figure 3).

For the Qwen models the two readouts move in opposite directions under post-training. In both families and at both temperatures, the post-trained model's single list is closer to the reference than the base model's (TV −0.043 to −0.086, all intervals exclude zero), covers more of its mass (+6.1 to +10.8 points), and names severe conditions more often (+9.1 to +13.4 points). Its sampled top answers move farther from the reference (+0.029 to +0.071), and five of them cover less (−1.9 to −6.0 points). Within each post-trained model the list beats its own samples by 0.04–0.09 in TV; within each base model the gap is about zero or favors the samples. The CoT prompt keeps the list's advantage (TV −0.042, severe naming +8.1 points) and removes the loss in the samples (+0.002, n.s.). This quantifies the observation of §4.2 that conditions dropping out of the top answer remain lower in the list.

At 32B the pattern holds in both families (Table 6): the post-trained lists are closer to the reference (TV −0.053 for Qwen2.5-32B-Instruct, −0.052 and −0.045 for the OLMo 3.1 DPO and Instruct models) and cover 5.6–6.7 points more of it, while the samples move away (+0.014 to +0.034). Within every post-trained 32B model the list beats its own samples by 0.04–0.09 in TV, against about zero for both base models. OLMo 3 7B is the exception with the five-item prompt. Its DPO and Instruct models' lists are no closer to the reference than the base model's (TV +0.000 and +0.028 at T=0.7; −0.026 and −0.005 at T=1) and cover at most 5 points more of it; only the naming of severe conditions improves (+6.1 to +12.8 points). Asked for a probability on every condition, its Instruct model's stated distribution is closer to the reference than the base model's (−0.032 [−0.038, −0.027]; §4.8), so the exception is specific to the short list. RL-Zero's lists improve slightly (TV −0.020 and −0.039). The medical descendants change little relative to their parent (TV −0.012 for AlphaMed, −0.014 for HuatuoGPT-o1, +0.017 for m1). In absolute terms every model is far from the reference: a single list covers 0.22–0.41 of the reference mass, against 0.74 for the reference's own top five.

Few-shot prompting. With three worked examples in every prompt, the Qwen gains persist or grow (TV −0.057 to −0.087; −0.055 to −0.073 without the true pathology), but OLMo 3.1 32B's list keeps only its higher reference mass (+0.047), with TV, TV without the true pathology and DDF1 no longer significant (Appendix 4.10). Averaging calls narrows the gap: the mean of five base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700); a post-trained list tracks its own base model's view of the patient and barely varies between calls (§4.18), and Qwen3-8B's thinking mode goes further, with one list beating five averaged base lists.

4.7 Consequences for sampling-based methods

We ask what the gap does to two common uses of repeated samples: aggregating them into an answer, and using their agreement as an uncertainty signal (Table 8). The single-call answer is the top answer of a model's first valid sample.

Aggregation. Majority voting over 32 samples (self-consistency; 41) raises base models' accuracy by 3.4–5.2 points at T=0.7 and 10.4–14.2 at T=1. For the general post-trained models it adds at most 0.8 points at T=0.7 (Qwen3-8B +0.3, Qwen2.5-7B-Instruct +0.8, OLMo-3-Instruct −0.2; all n.s.) and 0.1–4.4 at T=1. Models that reason before answering still benefit (HuatuoGPT-o1 +4.8,

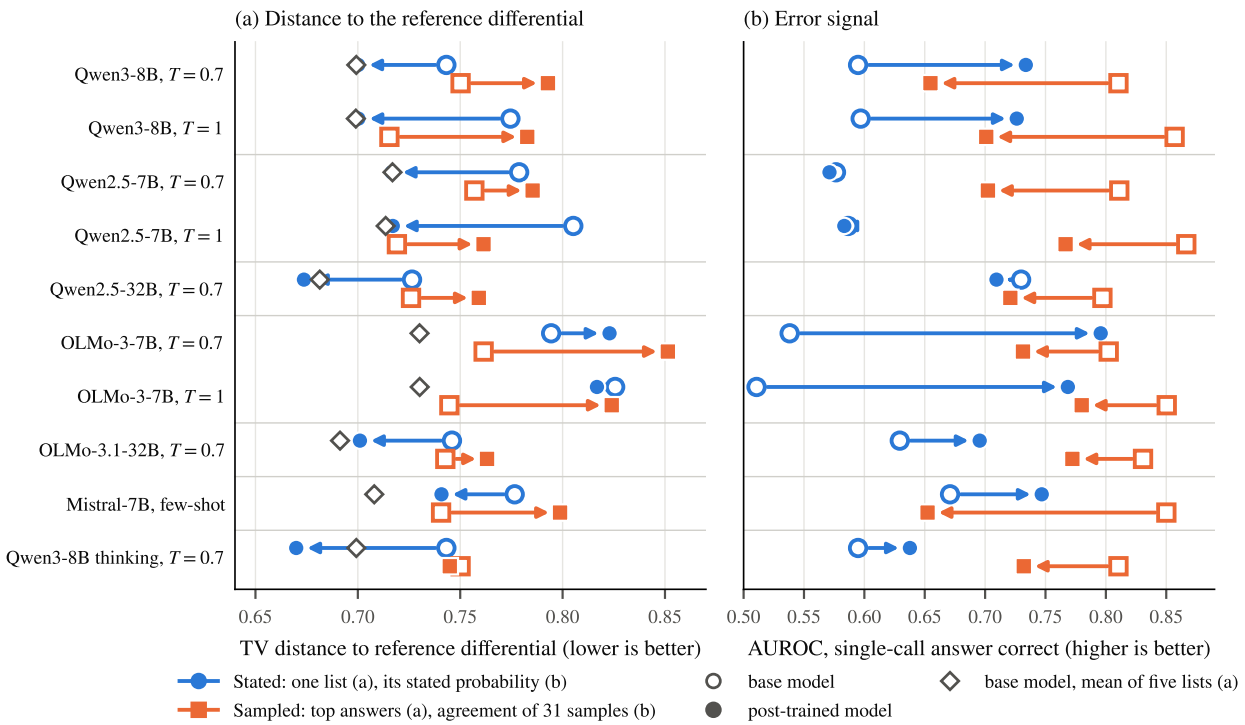


Figure 3: Base (hollow) → post-trained (filled) on DDXPlus (980 cases; OLMo: base → final Instruct; 32B models at T=0.7 only). (a) TV distance to the reference differential of one stated list (blue circles) and of the sampled top-1 answers (orange squares). (b) AUROC for the single-call answer being correct, of its stated probability (blue) and of agreement across 31 further samples (orange).

m1 +3.8, Qwen3-8B with the CoT prompt +4.5). Averaging the stated lists instead of voting changes accuracy by at most about 2 points for any model. For a top-three differential, the stated list is the better source after post-training. For Qwen3-8B, the top three of one list contains the true pathology 13.7 [+11.4, +15.9] points more often than the three most frequent answers in 32 samples (T=1: +10.0), and for Qwen2.5-7B-Instruct 7.3 [+5.3, +9.6] points more often (T=1: +2.0, n.s.). For every base model the 32 samples win, by 7.3–13.6 points at T=0.7. OLMo-3-Instruct is in between (+1.9, n.s.; T=1: −4.7). Collapsed samples often offer fewer than three answers, though: matched for size, the post-trained list is no better than its samples (−0.5 to −5.2 points), while base models' samples still win by 8.5–14.6 (§4.15).

Uncertainty. We score how well each signal ranks the correctness of the single-call answer (AUROC). The stated probability of that answer needs no extra calls; *agreement*, the share of the model's other samples with the same top answer, uses 4 or 31 more. For base models agreement is far better: 0.80–0.81 with 31 extra samples, against 0.54–0.60 for their nearly uninformative stated probability. Post-training weakens agreement in every family (Qwen3 0.810 → 0.655, Qwen2.5 0.811 → 0.702, OLMo 0.802 → 0.731), consistent with Liu [23], and moves the comparison toward the stated probability. For Qwen3-8B (0.734) and OLMo-3-Instruct (0.796), the stated probability is better than agreement over 32 samples (+0.079 [+0.046, +0.110] and +0.064 [+0.027, +0.101]); at T=1 the two are statistically tied. For Qwen2.5-7B-Instruct (0.571 vs. 0.702) and all three medical models, agreement remains better. These AUROCs are computed on each model's own

Pair	Model	TV to reference↓ One list	TV to reference↓ Samples	Reference mass covered One list	Reference mass covered 5 samples	Severe named One list
Qwen3-8B, T=0.7	base	0.743	**0.749**	0.335	**0.245**	0.347
	post-trained	**0.700**	0.793	**0.396**	0.207	**0.437**
Qwen3-8B, T=1	base	0.772	**0.712**	0.293	**0.275**	0.312
	post-trained	**0.700**	0.783	**0.396**	0.215	**0.440**
Qwen2.5-7B, T=0.7	base	0.779	**0.756**	0.291	**0.233**	0.345
	instruct	**0.716**	0.785	**0.364**	0.214	**0.440**
Qwen2.5-7B, T=1	base	0.803	**0.715**	0.255	**0.257**	0.304
	instruct	**0.717**	0.761	**0.363**	0.234	**0.438**
OLMo 3 7B, T=0.7	base	0.795	**0.757**	0.268	**0.237**	0.256
	SFT+DPO	0.795	0.804	0.277	0.193	**0.378**
	Instruct (RLVR)	0.823	0.851	0.257	0.150	0.317
	RL-Zero	**0.775**	0.793	**0.295**	0.197	0.274
Oracle	reference	0.265	0.18–0.23	0.735	0.563	0.975

Table 7: Stated vs. sampled differentials on DDXPlus (980 cases, valid samples only). *One list*: a single sample's stated differential, renormalized over the conditions it names (mean over samples). *Samples*: the empirical distribution of top-1 answers over the 32 samples; *5 samples*: reference mass of the distinct top answers in five random samples, the same five-name budget as one list. *Severe named*: share of (case, severe condition with reference probability ≥ 0.1) pairs that one list names. Bold marks the better model of each pair (OLMo: of the block). Oracle: the reference's own top five as a list, and draws from the reference as samples. Post-training improves the Qwen lists while worsening their samples; OLMo's lists do not improve.

answers, so they compare signals within a model, not models with each other.

Ambiguity. The reference differential also says how ambiguous each case is. We correlate each uncertainty signal with the entropy of the case's reference differential. For every model and signal the Spearman correlation lies between -0.10 and $+0.14$, except AlphaMed's stated probability and stated entropy ($+0.30$ and $+0.32$). How uncertain a model is about a case, by either readout, says almost nothing about how ambiguous the case is.

At 32B. The same shifts appear with smaller margins. Voting gains fall from 7.0 and 8.4 points for the base models to 1.9 and 0.9 (the latter n.s.) for the Instruct models, and agreement weakens as an error signal (Qwen 0.797 → 0.721, OLMo 0.831 → 0.772). For Qwen2.5-32B-Instruct the top three of one list beats the three most frequent of 32 sampled answers by 5.6 [3.6, 7.7] points, but its stated probability only ties agreement over 32 samples (-0.011, n.s.; it beats agreement over five, $+0.047$); for OLMo 3.1 32B Instruct agreement stays better (-0.077).

After post-training, then, repeated sampling adds little: voting no longer improves accuracy and agreement carries less signal, while one stated list carries as much of the differential or more. Where the crossover falls depends on the model, so neither readout can be assumed to be the informative one without checking.

4.8 Robustness of the stated readout to elicitation

The stated readout above uses a five-item list. We repeat the Qwen pairs and OLMo 3 7B's base and Instruct models at T=0.7 with a prompt that asks for a probability on every one of the 49 con-

Model	Single-call accuracy	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct		
				Stated p	Agree@5	Agree@32
Qwen3-8B-Base	0.456	+3.4	−7.3	0.595	0.772	**0.810**
Qwen3-8B	0.556	+0.3	+13.7	**0.734**	0.613	0.655
Qwen3-8B (CoT prompt)	0.648	+4.5	+0.0	0.718	0.725	**0.778**
Qwen2.5-7B	0.371	+5.2	−8.9	0.576	0.764	**0.811**
Qwen2.5-7B-Instruct	0.480	+0.8	+7.3	0.571	0.653	**0.702**
OLMo-3 base	0.291	+4.0	−13.6	0.538	0.761	**0.802**
OLMo-3 SFT+DPO	0.327	−0.4	−7.8	0.738	0.702	**0.751**
OLMo-3-Instruct	0.267	−0.2	+1.9	**0.796**	0.668	0.731
AlphaMed-7B	0.528	+0.9	+0.6	0.673	0.663	**0.715**
HuatuoGPT-o1-7B	0.532	+4.8	−0.4	0.590	0.715	**0.775**
m1-7B	0.495	+3.8	−4.6	0.680	0.736	**0.782**

Table 8: Aggregation and uncertainty on DDXPlus (T=0.7, 980 cases). Single-call answer: top-1 of the first valid sample. Vote@32 gain: accuracy of the majority top-1 over all valid samples minus single-call accuracy, in points. Top-3: rate at which one stated list's top three contains the true pathology, minus the rate for the three most frequent sampled answers (points). AUROC for the single-call answer being correct: its stated probability (no extra calls) vs. agreement of 4 or 31 further samples; bold marks the best signal. Post-trained models gain little from voting; for some of them the stated probability becomes the better error signal.

ditions, summing to one, and take each sample's top answer to be its highest-probability condition (Table 9). The results match the short list for Qwen. After post-training the stated distribution is closer to the reference (TV −0.052 for Qwen3-8B, −0.067 for Qwen2.5-7B-Instruct, −0.032 for OLMo-3-7B-Instruct) while the samples move away (+0.012 to +0.038); OLMo-3-7B-Instruct's full distribution moves closer although its five-item list does not. Qwen3-8B's stated probability again beats agreement over 32 samples (+0.069 [+0.033, +0.104]), Qwen2.5-7B-Instruct's again does not (−0.074), and OLMo-3-7B-Instruct's ties it (−0.010; +0.044 against five samples). The top three of one full distribution beats the three most frequent of 32 sampled answers for Qwen3-8B (+14.6 points) and narrowly for Qwen2.5-7B-Instruct (+1.6), but not for OLMo-3-7B-Instruct (−9.5).

4.9 Ruling out format, accuracy and floor effects

Table 11 tests simpler explanations of the stated-list result. For reference, one list's recall@5 rises after post-training by 13.6, 16.2 and 12.0 points for Qwen3-8B, Qwen2.5-7B and Qwen2.5-32B at T=0.7 and by 21.9 and 24.0 for the 7–8B Qwen models at T=1, by 12.7 for OLMo 3.1 32B, and for OLMo 3 7B by 1.5 (n.s.) at T=0.7 and 10.0 at T=1 (condition-clustered intervals exclude zero except where marked). *Format.* The general models' post-trained lists and base-model lists at T=0.7 are nearly all complete (81–99% name five on-list conditions with probabilities; Qwen base models at T=1, 61–62%; the medical models, 61–80%). Restricting to complete lists leaves the gaps unchanged (Qwen2.5-7B: 0.779 vs. 0.719). *Accuracy.* A more accurate model puts more stated mass on the true pathology, which the reference also favors. Within samples whose top answer is correct, and within those whose top answer is wrong, the post-trained Qwen list is still closer to the reference (−0.018 to −0.034 and −0.036 to −0.079 in TV across both temperatures; Table 11 shows T=0.7). *Floor.* One list can do no better than the reference's own top five, and a finite set of samples no better than as many draws from the reference. Measured as excess over these floors, post-training lowers the list's TV (Qwen3-8B 0.479 → 0.435) and raises the samples'

Model	TV to reference↓ One distribution	Samples	AUROC, single-call answer correct Stated p	Agree@5	Agree@32	Top-3: one dist. − 32 samples
Qwen3-8B-Base	0.733	**0.752**	0.647	0.809	**0.845**	−6.7
Qwen3-8B	**0.681**	0.790	**0.717**	0.609	0.647	+14.6
Qwen2.5-7B	0.769	**0.758**	0.688	0.792	**0.827**	−10.2
Qwen2.5-7B-Instruct	**0.702**	0.770	0.674	0.715	**0.748**	+1.6
OLMo-3-1025-7B	0.807	**0.766**	0.610	0.744	**0.786**	−15.3
OLMo-3-7B-Instruct	**0.775**	0.787	0.749	0.705	**0.759**	−9.5

Table 9: Full-distribution prompt on DDXPlus (T=0.7, 980 cases): a probability on every condition, top answer = highest stated probability. TV: one sample's stated distribution vs. the sampled top-1 answers (bold: better model of each pair). AUROC: bold marks each model's best signal.

(0.557 → 0.613). *True pathology removed.* With the true pathology removed from both list and reference and both renormalized, the post-trained Qwen and OLMo 3.1 32B lists are still closer to the reference (−0.018 to −0.056), and DDXPlus's set F1 against the reference's support improves by 0.040–0.083 (Table 34). TV, mass and DDF1 are not independent, since a list's TV is the reference mass it leaves out plus the mass it misallocates among the conditions it names, and the second term grows mechanically with the first. The conditional TV, between the list and the reference restricted to the named conditions and renormalized, removes this: it falls after post-training for every Qwen pair and OLMo 3.1 32B (−0.025 to −0.078, all surviving correction) and rises for OLMo 3 7B at T=1 (+0.064), so the Qwen lists both name better conditions and weight them more like the reference. *Patient-blind list.* A fixed list of the five conditions with the highest average reference probability scores TV 0.794, mass 0.238 and DDF1 0.272. Every model's list beats it in TV except OLMo 3 7B's and Qwen2.5-7B's base list at T=1; in DDF1 several base lists do not (Qwen2.5-7B 0.245, OLMo 3 7B 0.225), and the post-trained Qwen and OLMo 3.1 32B lists exceed it by 0.035–0.061. *Over-naming.* As a share of named conditions, post-trained lists name fewer that the reference gives less than 1% (Qwen3-8B 0.59 → 0.55) and fewer unwarranted severe conditions (Qwen2.5-32B 0.74 → 0.65), but as a count per list the change is mixed: from 0.22 fewer unwarranted severe conditions (OLMo 3.1 32B) to 0.35 more (OLMo 3 7B). *Severe conditions one by one.* Table 10 tests each of the 17 severe conditions, counting it wherever the reference gives it any weight. After Benjamini–Hochberg correction, 2–7 conditions per zero-shot pair are named significantly less often after post-training and 0–6 significantly more often, and no pair names more conditions less often than a random swap of base and post-trained lists would. With the warranted threshold at 0.01 or 0.05 instead of 0.1, post-trained lists still name more severe conditions in seven of eight zero-shot settings (+2.5 to +8.1 points; OLMo 3.1 32B n.s.). That count test is weak against an overall upward shift, so Table 10 also gives the mean shift across the 17 conditions: it rises significantly in seven of eight zero-shot settings but falls with worked examples for Qwen2.5-32B (−0.012) and OLMo 3 7B (−0.018), and is flat for Mistral (−0.009, p=0.05). Significant losses include acute pulmonary edema (Qwen3-8B 0.75 → 0.61, Qwen2.5-32B 0.63 → 0.44, OLMo 3 7B 0.56 → 0.31, OLMo 3.1 32B 0.57 → 0.41), stable angina (Qwen2.5-7B 0.25 → 0.08), NSTEMI/STEMI (OLMo 3 7B 0.31 → 0.14, OLMo 3.1 32B 0.49 → 0.36) and myocarditis (Qwen2.5-32B 0.14 → 0.05); significant gains include pulmonary embolism (Qwen2.5-7B 0.29 → 0.56, OLMo 3 7B 0.20 → 0.65), NSTEMI/STEMI (Qwen2.5-7B 0.32 → 0.52) and unstable angina (Qwen3-8B 0.14 → 0.37). Among single stages, OLMo 3 7B's RLVR step and OLMo 3.1 32B's SFT and RLVR steps do name more conditions less often than chance (12–13 of 17, p=0.03–0.04).

Sampling from the list. Replacing each sample's top answer with a draw from its own stated list

Pair (base → post)	Named less / more often	Significant less / more	P(as many less often)	Mean shift (17 conditions)	P	Largest loss (shift)
Qwen3-8B, T=0.7	5 / 12	2 / 3	0.98	+0.040	0.001	acute pulmonary edema (-0.14)
Qwen3-8B, T=1	6 / 11	2 / 4	0.94	+0.060	0.001	stable angina (-0.06)
Qwen2.5-7B, T=0.7	5 / 12	2 / 2	0.98	+0.029	0.001	stable angina (-0.16)
Qwen2.5-7B, T=1	4 / 13	2 / 5	0.99	+0.065	0.001	stable angina (-0.13)
Qwen2.5-32B, T=0.7	6 / 11	3 / 5	0.93	+0.047	0.001	acute pulmonary edema (-0.19)
OLMo 3 7B, T=0.7	8 / 9	7 / 6	0.68	+0.046	0.001	acute pulmonary edema (-0.24)
OLMo 3 7B, T=1	8 / 9	6 / 6	0.68	+0.061	0.001	NSTEMI/STEMI (-0.12)
OLMo 3.1 32B, T=0.7	7 / 10	5 / 0	0.82	+0.001	0.876	acute pulmonary edema (-0.16)
Qwen3-8B, few-shot	5 / 12	1 / 4	0.97	+0.061	0.001	acute pulmonary edema (-0.16)
Qwen2.5-7B, few-shot	8 / 9	4 / 2	0.69	+0.034	0.001	stable angina (-0.15)
Qwen2.5-32B, few-shot	10 / 7	5 / 3	0.29	−0.012	0.009	laryngospasm (-0.25)
OLMo 3 7B, few-shot	9 / 8	7 / 3	0.51	−0.018	0.018	croup (-0.51)
OLMo 3.1 32B, few-shot	9 / 8	4 / 2	0.50	+0.011	0.012	stable angina (-0.25)
Mistral-7B, few-shot	11 / 6	3 / 0	0.17	−0.009	0.052	pulmonary embolism (-0.31)

Table 10: The 17 severe conditions one by one (DDXPlus). For each condition, the share of a model's lists that name it, over the cases whose reference gives it any weight; post-trained minus base, with intervals that resample the true pathologies (50,000 resamples for the zero-shot rows, 10,000 for the few-shot rows) and Benjamini–Hochberg correction across the 17 conditions. P columns compare the count of conditions named less often and the mean shift across conditions with a null that swaps each case's base and post-trained lists at random (2,000 swaps); the count test is weak against an overall upward shift, so the mean shift and the largest single loss are shown too.

raises pass@32 for every model. Compared like for like, post-trained lists reach higher than their base model's lists in five of seven comparisons (Qwen3-8B 0.811 vs. 0.780 at T=0.7, Qwen2.5-32B-Instruct 0.880 vs. 0.850, OLMo 3.1 32B Instruct 0.853 vs. 0.839), but not for Qwen3-8B at T=1 (0.820 vs. 0.832) or OLMo 3 7B (0.577 vs. 0.719), and a uniform guesser reaches 0.48. OLMo 3 7B remains the exception throughout: its Instruct list is closer to the reference when its top answer is correct but much farther when it is wrong.

4.10 Few-shot elicitation

The main analyses give every model the same zero-shot chat prompt, which base models were not trained to follow; they give invalid answers on up to 47% of samples, which we count as wrong. We therefore re-ran the five final base/post-trained pairs at T=0.7 with three worked examples in the prompt: three fixed DDXPlus training patients with different true pathologies, each followed by the top five conditions of its reference differential, with probabilities, in the JSON format the instructions ask for. Base models see the instructions and examples as a plain document and complete the next one (generation stops at the end of the list); post-trained models see the same instructions and examples as earlier chat turns. Invalid samples fall below 0.5% for every model (Table 12).

The examples help base models most. Their pass@1 rises from 0.178 to 0.306 for OLMo-3-1025-7B, from 0.326 to 0.528 for OLMo-3-1125-32B and from 0.500 to 0.647 for Qwen2.5-32B, and their pass@32 reaches 0.76–0.90, while their stated lists barely change (TV 0.712–0.787, against 0.726–0.794 zero-shot). The losses in the samples therefore grow (Table 13): post-training costs 15.2–29.9 points of pass@32 in every pair, including both 32B pairs, whose zero-shot losses did not survive correction, and every post-trained model's Cov@32 (0.21–0.25) is below every base model's (0.33–0.45) and below a uniform guesser's.

Model (T=0.7)	Complete lists	TV of one list↓ complete only	top-1 right	top-1 wrong	Excess TV over floor↓ One list	Samples	Severe named ref. < 0.1	pass@32 list-sampled
Qwen3-8B-Base	0.90	0.743	0.663	0.806	0.479	**0.557**	0.752	0.780
Qwen3-8B	0.99	**0.701**	**0.646**	**0.769**	**0.435**	0.613	**0.680**	**0.811**
Qwen2.5-7B	0.95	0.779	0.689	0.828	0.514	**0.575**	0.823	0.772
Qwen2.5-7B-Instruct	0.91	**0.719**	**0.655**	**0.773**	**0.452**	0.605	**0.796**	**0.802**
Qwen2.5-32B	0.96	0.725	0.661	0.795	0.462	**0.543**	0.735	0.850
Qwen2.5-32B-Instruct	0.98	**0.674**	**0.639**	**0.726**	**0.409**	0.579	**0.645**	**0.880**
OLMo-3-1025-7B	0.81	**0.794**	0.687	**0.837**	**0.524**	**0.522**	**0.747**	**0.719**
OLMo-3-7B-Instruct	0.95	0.823	**0.659**	0.882	0.558	0.671	0.779	0.577
OLMo-3-1125-32B	0.89	0.744	0.662	0.812	0.481	**0.532**	0.706	0.839
OLMo-3.1-32B-Instruct	0.99	**0.701**	**0.644**	**0.774**	**0.436**	0.583	**0.634**	**0.853**

Table 11: Robustness of the stated-list result (DDXPlus, T=0.7, 980 cases, valid samples). Complete lists: share naming five on-list conditions with valid probabilities. TV of one list on complete lists only, and on samples whose top answer is right or wrong. Excess TV: TV minus the expected TV of the matching oracle (the reference's top five for one list; as many reference draws as the model has valid samples for the samples). Severe named, ref. < 0.1: share of named severe conditions that the reference gives less than 0.1 (over-naming). pass@32 list-sampled: top answers drawn from each sample's stated list. Bold: better model of each pair. Excess TV is averaged over samples against each case's floor, so it is not exactly the difference of the mean TV and the mean floor. A uniform guesser scores about 0.48 at pass@32.

For the three Qwen pairs the stated-list improvement persists and mostly grows: TV −0.057 to −0.087, −0.055 to −0.073 with the true pathology removed, and DDF1 +0.051 to +0.081, all surviving correction. OLMo 3 7B remains the exception (TV +0.021, n.s.; +0.024 without the true pathology). For OLMo 3.1 32B the improvement shrinks: the post-trained list still places more reference mass on the conditions it names (+0.047), but its TV (−0.022), TV without the true pathology (−0.006) and DDF1 (+0.009) no longer survive correction. At 32B the stated-list result is therefore robust for Qwen2.5-32B but depends on the prompt for OLMo 3.1 32B. Post-trained lists still name 5–10 of the 17 severe conditions less often than their base model's.

The consequences replicate in all five pairs. Majority voting over 32 samples gains base models 6.2–10.9 points and post-trained models 0.2–1.1. The three most frequent of 32 sampled answers beat the top three of one stated list by 8.2–22.0 points for base models, while for post-trained models the list is ahead by 0.2–16.7. Agreement with four more samples is the better error signal for every base model (AUROC 0.79–0.84, against 0.67–0.73 for the stated probability), and the stated probability for every post-trained model (0.69–0.77, against 0.56–0.68); against agreement across 31 samples the stated probability wins clearly only for Qwen3-8B and Qwen2.5-7B-Instruct (+0.108 and +0.130, clustered) and ties for the others. This check uses one temperature and one set of examples, and the examples show the reference's own lists, which may help every model match the reference.

4.11 Two further controls

Reference-free few-shot examples. The examples of §4.10 show each training patient's reference top five, and so also the target of the evaluation. We re-ran the six final pairs with the same three patients but with each example list replaced by Qwen2.5-32B-Instruct's own greedy answer (on-list, with the true pathology first for two patients and second for one). The conclusions do not change (Table 14). Post-training costs 12.7–29.7 points of pass@32 in every pair. The stated list moves closer to the reference for the three Qwen pairs and, now clearly, for Mistral (TV −0.053 to −0.077; −0.031 to −0.059 with the true pathology removed), but not for OLMo 3 7B or OLMo 3.1

Model	Invalid (%)	pass@1	pass@32	Cov@32	TV, one list all	TV, one list w/o true	DDF1	TV samples	Vote@32 gain	Top-3 list − samples	AUROC stated	AUROC agree@5	AUROC agree@32
Qwen3-8B-Base	0.1	0.403	**0.809**	**0.383**	0.768	0.860	0.245	**0.732**	+8.4	−16.0	0.722	0.828	0.862
Qwen3-8B	0.0	**0.601**	0.639	0.221	**0.682**	**0.805**	**0.327**	0.786	+0.2	+14.5	0.693	0.563	0.584
Qwen2.5-7B (base)	0.4	0.398	**0.761**	**0.333**	0.784	0.879	0.238	**0.756**	+6.2	−15.6	0.734	0.792	0.827
Qwen2.5-7B-Instruct	0.0	**0.450**	0.535	0.220	**0.702**	**0.806**	**0.309**	0.798	+0.6	+16.7	0.757	0.594	0.626
Qwen2.5-32B (base)	0.1	0.647	**0.896**	**0.343**	0.712	0.847	0.298	**0.715**	+7.1	−8.2	0.726	0.838	0.875
Qwen2.5-32B-Instruct	0.0	**0.663**	0.744	0.252	**0.654**	**0.775**	**0.349**	0.758	+0.6	+10.1	0.718	0.647	0.694
OLMo-3-1025-7B (base)	0.2	0.306	**0.857**	**0.454**	**0.787**	**0.859**	**0.245**	**0.712**	+10.9	−22.0	0.674	0.823	0.871
OLMo-3-7B-Instruct	0.1	**0.338**	0.558	0.213	0.808	0.884	0.238	0.830	+0.7	+0.2	0.733	0.668	0.696
OLMo-3-1125-32B (base)	0.4	0.528	**0.888**	**0.397**	0.719	0.818	0.302	**0.699**	+8.9	−13.4	0.670	0.826	0.865
OLMo-3.1-32B-Instruct	0.0	**0.575**	0.697	0.249	**0.697**	**0.812**	**0.311**	0.772	+1.1	+3.7	0.772	0.684	0.744

Table 12: Few-shot elicitation on DDXPlus (T=0.7, 980 cases, 32 samples; three worked examples in the prompt for every model, §4.10). Invalid: samples naming no listed condition. TV: distance to the reference differential, for one stated list (with the true pathology removed from both in "w/o true") and for the sampled top answers. Vote@32 gain and top-3 list minus top-3 of the samples are in accuracy points. AUROC: for the single-call answer being correct, of its stated probability and of agreement with the next 4 or the next 31 samples. Bold: better of the pair.

Pair (base → post)	pass@32 (points)	TV samples	TV, one list all	TV, one list w/o true	DDF1	Reference mass of one list
Qwen3-8B	−12.4 / −17.0	+0.043 / +0.054	−0.043 / −0.087	−0.018† / −0.055	+0.042 / +0.081	+0.062 / +0.113
Qwen2.5-7B	−9.7 / −22.7	+0.028† / +0.042	−0.062 / −0.082	−0.041 / −0.073	+0.062 / +0.072	+0.073 / +0.124
Qwen2.5-32B	−4.9† / −15.2	+0.033 / +0.044	−0.053 / −0.057	−0.043 / −0.072	+0.051 / +0.051	+0.067 / +0.084
OLMo 3 7B	−28.3 / −29.9	+0.090 / +0.118	+0.029† / +0.021†	+0.034 / +0.024	−0.006† / −0.007†	−0.010† / −0.002†
OLMo 3.1 32B	−4.0† / −19.1	+0.020† / +0.074	−0.045 / −0.022†	−0.023 / −0.006†	+0.040 / +0.009†	+0.056 / +0.047

Table 13: Base → post-trained differences under the zero-shot prompt / the few-shot prompt (DDXPlus, T=0.7). †: not significant under the condition-clustered bootstrap with Holm correction (over 112 tests for the zero-shot and 70 for the few-shot pairs; Appendix 7).

32B, and wherever the list improves its allocation among the named conditions also moves closer to the reference (conditional TV −0.051 to −0.100). Voting helps base models by 3.7–13.7 points and post-trained ones by at most 1.4; the top three of one list beats the three most frequent samples for every post-trained model; the stated probability beats agreement as an error signal for every post-trained model and agreement beats it for every base model; and no pair names more severe conditions less often than chance. Five of six base models score lower at k=1 than with reference-derived examples (OLMo-3-1125-32B 0.453 against 0.528).

Matched temperature. Post-trained models are sharper than their base models at the same nominal temperature, so part of the narrowing could be colder effective sampling. We sampled Qwen3-8B and Qwen2.5-7B-Instruct at T=1.3 and 1.6 (Table 15), and the two models differ. Qwen3-8B barely broadens: at T=1.6 it gives 1.92 distinct answers and reaches 0.684 at pass@32, still below its base model at T=0.7 (0.732) and at T=1 (0.789; −10.5 points, surviving correction); voting gains stay under 2 points, agreement stays weaker than for the base model, and its stated list does not change (TV 0.701). Qwen2.5-7B-Instruct does broaden: at T=1.3, with 2% invalid answers, it gives 3.38 distinct answers, matches its base model's pass@32 at T=0.7 (0.704 against 0.695), gains 5.6 points from voting, and agreement is again the better error signal (0.794 against 0.602), at a cost of 4.2 points of pass@1; at T=1.6, 31% of its answers are invalid. Its stated list stays closer to the reference than the base model's (0.722 against 0.779). For some models, then, much of the sampled narrowing is sharper sampling that a higher temperature undoes; the stated-list result is not.

Pair (base → post)	pass@32 (points)	TV samples	TV, one list all	TV, one list w/o true	DDF1	Cond. TV
Qwen3-8B	−14.6	+0.049	−0.077	−0.049	+0.079	−0.078
Qwen2.5-7B	−19.7	+0.031†	−0.077	−0.059	+0.074	−0.094
Qwen2.5-32B	−14.5	+0.036	−0.053	−0.057	+0.054	−0.051
OLMo 3 7B	−29.7	+0.116	−0.007†	+0.013†	−0.002†	−0.015†
OLMo 3.1 32B	−17.4	+0.073	−0.016†	+0.002†	+0.009†	−0.005†
Mistral-7B	−12.7	+0.038†	−0.059	−0.031	+0.057	−0.100

Table 14: Few-shot control with reference-free examples (DDXPlus, T=0.7, 980 cases): the same three training patients as in §4.10, but each example list is Qwen2.5-32B-Instruct's own greedy answer rather than the reference's top five. Post-trained minus base. †: not significant (condition-clustered bootstrap, Holm over 84 tests; conditional TV Holm-corrected as a separate family).

Model	T	Invalid (%)	pass@1	pass@32	Distinct	TV samples	TV one list	Vote@32 gain	AUROC stated	AUROC agree@32
Qwen3-8B-Base	0.7	11.3	0.388	0.732	3.77	0.750	0.743	+3.4	0.595	0.810
	1	22.3	0.282	0.789	6.60	0.715	0.775	+10.4	0.597	0.857
Qwen3-8B	0.7	0.0	0.559	0.607	1.27	0.793	0.700	+0.3	0.734	0.655
	1	0.0	0.556	0.643	1.53	0.783	0.700	+1.4	0.726	0.701
	1.3	0.0	0.552	0.668	1.73	0.776	0.701	+1.0	0.707	0.724
	1.6	0.0	0.549	0.684	1.92	0.770	0.701	+1.8	0.715	0.745
Qwen2.5-7B	0.7	0.9	0.352	0.695	4.91	0.757	0.779	+5.2	0.576	0.811
	1	18.4	0.239	0.766	8.23	0.719	0.805	+14.1	0.587	0.867
Qwen2.5-7B-Instruct	0.7	0.0	0.478	0.598	1.82	0.785	0.717	+0.8	0.571	0.702
	1	0.0	0.464	0.663	2.65	0.761	0.717	+4.4	0.583	0.767
	1.3	2.2	0.436	0.704	3.38	0.745	0.722	+5.6	0.602	0.794
	1.6	30.5	0.304	0.726	3.88	0.729	0.749	+10.0	0.637	0.814

Table 15: Matched-temperature control (DDXPlus, 980 cases, 32 samples, zero-shot prompt): the post-trained models sampled at higher temperatures (top-p 1) against their base models at T=0.7 and T=1. Columns as in Tables 2, 7 and 8.

4.12 Greedy decoding, temperature frontier and a must-not-miss prompt

Greedy single call. Deployed systems often decode greedily. We compared one greedy answer per case under three prompt pairings (Table 16): both arms with the reference-derived worked examples of §4.10 (*same examples*); base models with the reference-free examples of Table 14 against zero-shot post-trained models, so that neither arm sees a reference list (*reference-free base*); and reference-example base models against zero-shot post-trained models, the pairing we first ran. Under every pairing post-training significantly raises top-1 in at most two of six pairs. With the same examples these are Qwen3-8B (+13.7) and Mistral (+10.8); Qwen2.5-7B gains +1.8 and the other three lose 4.8–5.8 points, all n.s. With reference-free base models only Qwen3-8B gains (+10.7), while Qwen2.5-32B (−8.0) and OLMo 3 7B (−14.5) lose significantly. The example lists matter little to the base models (top-1 within 2.7 points, except OLMo 3.1 32B, 0.630 against 0.559) but raise five of the six post-trained models (Mistral 0.417 to 0.577, OLMo 3 7B 0.258 to 0.345; Qwen2.5-7B-Instruct falls to 0.458). Under every pairing the two smaller Qwen models' lists contain the true pathology more often (recall@5 +16.8 to +26.0) and name more warranted severe conditions, every Qwen list is closer to the reference, and OLMo 3 7B's lists are farther. Together with the few-shot comparison, where the sampled pass@1 gain is already only 1.6–5.2 points in four of six pairs, most of post-training's advantage at one sampled call for the 7–8B Qwen models

(+12.6 to +27.5 points at T=0.7 and 1, §4.2) is therefore prompt format and the base model's sampling noise; Qwen3-8B and Mistral keep a real gain.

Temperature frontier. With worked examples and top-p fixed at 1, we sampled the three 7–8B pairs at $T = 0.4$, 0.7, 1.0 and 1.3 (Table 17). Post-trained pass@1 barely moves with temperature (Qwen3-8B 0.60, Qwen2.5-7B-Instruct 0.43–0.46, OLMo-3-7B-Instruct 0.29–0.35), pass@32 rises (to 0.70, 0.66 and 0.74 at T=1.3), and the stated list barely changes (TV 0.68, 0.70 and 0.81–0.83), consistent with the concentration described by Yang et al. [45]. Base models trade pass@1 (Qwen3-8B-Base 0.45 to 0.18) for pass@32, which peaks at T=1 or 1.3 (0.85–0.88), and 25–36% of their answers are invalid at T=1.3. The frontiers differ by family. For Qwen3-8B the post-trained model is better at pass@1 and the base model at pass@32 at every temperature tested; for Qwen2.5-7B the curves meet (at pass@1 about 0.43, the base model at T=0.4 and the Instruct model at T=1.3 both reach pass@32 0.66); and for OLMo 3 7B the base model at T=0.4 beats its Instruct model at every temperature on both (0.368 and 0.803, against at best 0.345 and 0.737). Agreement becomes a better error signal for post-trained models as temperature rises (Qwen3-8B 0.57 to 0.68) but stays below the base models' (0.81–0.92).

Must-not-miss prompt. The standard prompt asks for the most likely conditions. We asked the six post-trained models for the five most likely conditions and then up to three dangerous conditions to rule out (Table 18). Whole lists then name 0.44–0.58 of the severe conditions that the reference weights at least 0.1, 6–15 points more than with the standard prompt and 11–21 more than the base models' lists, and acute pulmonary edema is recovered for every model but Qwen3-8B (e.g. Qwen2.5-32B 0.44 → 0.63, OLMo 3 7B 0.31 → 0.58). The cost is 1.1–2.1 more unwarranted severe conditions per list, mostly pulmonary embolism, named in 88–100% of eligible cases, and for Qwen3-8B, Qwen2.5-7B-Instruct and Mistral a shorter most-likely part that contains the true pathology 5–10 points less often. Even when asked, no model names more than 0.58 of the warranted severe conditions.

4.13 Must-not-miss conditions defined outside DDXPlus

The severity label used above is the dataset's own. We therefore re-ran every severe-condition analysis with two published sets of conditions that must not be missed, mapped onto DDXPlus's 49 (Table 19): a harm-based set, the DDXPlus conditions among the 15 diseases that cause about half of serious misdiagnosis-related harms in the USA [29] (NSTEMI/STEMI, pulmonary embolism, pneumonia and pulmonary neoplasm, of which DDXPlus labels only the first two severe), and the life-threatening causes of acute chest pain in the 2021 AHA/ACC guideline [10] (acute coronary syndrome as NSTEMI/STEMI and unstable angina, pulmonary embolism, esophageal rupture as Boerhaave, and pneumothorax). Their union has seven conditions. Neither set involves a judgment of these vignettes, and both were fixed before the analysis. On average the result is stronger than with DDXPlus's labels. Zero-shot, post-trained lists name 9–24 points more of the warranted union conditions in seven of eight settings (OLMo 3.1 32B −0.3, n.s.), and with worked examples 7–22 points more for the three Qwen models and OLMo 3 7B (OLMo 3.1 32B +6.5 and Mistral +4.0, n.s.); the harm-based and chest-pain sets give the same zero-shot pattern. The losses of specific conditions, however, are confined to OLMo, Mistral and the few-shot Qwen2.5 models. No Qwen model names any union condition significantly less often zero-shot, while OLMo 3 7B loses NSTEMI/STEMI and unstable angina at both temperatures (and pulmonary neoplasm at T=1), OLMo 3.1 32B loses NSTEMI/STEMI, pulmonary embolism and pneumothorax, and with worked examples Mistral loses four of the seven (pulmonary embolism 31 points), Qwen2.5-7B-Instruct

Pair	Top-1 accuracy: base, examples reference	base, examples ref.-free	post-trained zero-shot	post-trained examples	Top-1, post − base (points): same examples	ref.-free base	original
Qwen3-8B	0.471	0.450	0.558	0.607	+13.7	+10.7	+8.7†
Qwen2.5-7B	0.437	0.420	0.496	0.458	+1.8†	+7.6†	+5.7†
Qwen2.5-32B	0.712	0.686	0.606	0.664	−4.8†	−8.0	−10.6
OLMo 3 7B	0.403	0.403	0.258	0.345	−5.8†	−14.5	−14.5
OLMo 3.1 32B	0.630	0.559	0.558	0.571	−5.7†	−0.1†	−7.1†
Mistral-7B	0.468	0.446	0.417	0.577	+10.8	−2.9†	−5.1†

Pair	Recall@5, post − base (points): same	ref.-free	original	TV one list, post − base: same	ref.-free	original	Severe named, post − base: same	ref.-free	original
Qwen3-8B	+26.0	+21.6	+24.2	−0.108	−0.077	−0.088	+0.127	+0.127	+0.127
Qwen2.5-7B	+20.1	+18.3	+16.8	−0.074	−0.075	−0.060	+0.091	+0.119	+0.110
Qwen2.5-32B	+0.3†	−2.4†	−5.3†	−0.054	−0.035	−0.031	+0.058	+0.046	+0.062
OLMo 3 7B	+2.0†	−17.2	−14.1	+0.060	+0.067	+0.083	+0.004†	−0.058†	−0.020†
OLMo 3.1 32B	+2.5†	−0.5†	−2.0†	+0.004†	+0.004†	+0.012†	−0.005†	−0.033†	−0.042†
Mistral-7B	+16.7	+2.9†	+4.1†	−0.009†	+0.008†	+0.029	−0.034†	−0.034†	−0.059†

Table 16: Greedy single call (DDXPlus, 980 cases, temperature 0, one answer per case). Base models are prompted with the worked examples of §4.10, whose lists are the reference top five ("reference") or Qwen2.5-32B-Instruct's answers ("ref.-free"); post-trained models are prompted zero-shot or with the reference examples. Three comparisons: *same examples* (both arms see the reference examples), *ref.-free base* (reference-free base against zero-shot post-trained, so neither arm sees a reference list), and *original* (reference-example base against zero-shot post-trained). Differences are over cases where both models gave a valid answer, so they can differ slightly from the difference of the level columns. †: the condition-clustered 95% interval includes zero.

two and Qwen2.5-32B-Instruct one (Boerhaave). When the true pathology is itself in the union set (140 cases), zero-shot post-trained lists contain it among their top five 10–26 points more often for four of five pairs (OLMo 3 7B +9.9, n.s.). Scored against the union set, the must-not-miss prompt names 15–26 points more of the warranted conditions than the standard prompt and 8–20 points more than two combined standard lists, a larger share of its gain than under DDXPlus's labels, at a cost of 0.7–1.5 more unwarranted union conditions per list.

4.14 Chance-corrected pass@32 and other models' views

Chance. With 49 listed conditions, part of pass@32 comes from answers that do not depend on the patient. We therefore also score each case's samples against the true pathology of a random case with a different pathology (Table 20). Base models' samples then still reach 0.05–0.19 at pass@32 and post-trained ones 0.01–0.04, far below the 0.48 of a uniform guesser, because both concentrate on conditions that fit the presentation. Subtracting these values shrinks the loss but keeps it in every pair with worked examples: 6.7–17.8 points with the reference-derived examples and 4.1–14.3 with another model's. Zero-shot, the corrected loss is 7.5 and 3.6 points for the 7–8B Qwen models and 25.2 for OLMo 3 7B, and it disappears at 32B (−0.5 and +0.1).

Other models' views. The packing test of Table 24 compares a post-trained list with its base model's mean for the same patient against the base model's means for other patients with the same diagnosis, a lenient baseline that any list responding to the patient passes. Against other families'

Model	T	Invalid (%)	pass@1	pass@32	Distinct	TV one list	AUROC stated	AUROC agree@32
Qwen3-8B-Base	0.4	0.1	0.447	0.727	3.64	0.772	0.687	0.806
	0.7	0.3	0.388	0.832	7.09	0.774	0.708	0.876
	1.0	3.0	0.304	0.880	10.33	0.800	0.703	0.909
	1.3	29.5	0.179	0.857	10.39	0.850	0.630	0.921
Qwen3-8B	0.4	0.0	0.602	0.632	1.14	0.681	0.693	0.569
	0.7	0.0	0.601	0.655	1.25	0.682	0.693	0.613
	1.0	0.0	0.599	0.676	1.40	0.682	0.695	0.651
	1.3	0.0	0.596	0.703	1.57	0.682	0.694	0.679
Qwen2.5-7B	0.4	0.4	0.421	0.662	3.13	0.778	0.748	0.805
	0.7	0.6	0.388	0.784	5.72	0.790	0.719	0.843
	1.0	2.6	0.333	0.841	8.53	0.815	0.692	0.899
	1.3	25.3	0.220	0.848	9.24	0.857	0.615	0.916
Qwen2.5-7B-Instruct	0.4	0.0	0.455	0.516	1.40	0.702	0.750	0.613
	0.7	0.0	0.448	0.556	1.78	0.703	0.747	0.643
	1.0	0.0	0.441	0.597	2.28	0.704	0.741	0.687
	1.3	0.2	0.430	0.660	2.98	0.707	0.720	0.729
OLMo-3-1025-7B	0.4	0.1	0.368	0.803	5.46	0.762	0.667	0.815
	0.7	0.4	0.294	0.865	9.62	0.795	0.663	0.879
	1.0	2.8	0.214	0.881	12.40	0.834	0.618	0.899
	1.3	36.2	0.110	0.801	10.69	0.888	0.552	0.882
OLMo-3-7B-Instruct	0.4	0.1	0.345	0.504	1.99	0.809	0.737	0.663
	0.7	0.1	0.335	0.602	2.95	0.808	0.740	0.716
	1.0	0.1	0.324	0.668	4.09	0.809	0.744	0.760
	1.3	5.6	0.286	0.737	5.88	0.831	0.680	0.775

Table 17: Temperature frontier with the few-shot prompt (DDXPlus, 980 cases, 32 samples, top-p 1 at every temperature).

base models alone, every zero-shot post-trained list is closer to its own base model's mean (-0.031 to -0.138). Table 21 compares it with every other final model's mean for the same patient. For Qwen3-8B, Qwen2.5-32B and OLMo 3.1 32B the post-trained list is closer to its own base model's view than to any other base model's (by 0.036–0.058) and far closer than to the average post-trained model's (0.067–0.090), but about as close to the nearest other post-trained model's (n.s.). Qwen2.5-7B-Instruct is as close to Qwen3-8B-Base's view as to its own base model's, and OLMo-3-7B-Instruct is closer to Qwen3-8B-Base's and Qwen3-8B's. For the stronger models, then, a post-trained list stays close to its own base model's view, while the stronger post-trained models also converge on one another.

4.15 Ties, list size, validity, prevalence and the reference ceiling

These checks use the zero-shot T=0.7 final pairs. *Ties.* Post-trained models' 32 samples are unanimous in 34–57% of cases (Qwen3-8B 78%), where agreement cannot rank anything. On the remaining cases agreement's AUROC is 0.61–0.66 for the post-trained models against 0.78–0.83 for their base models, so agreement weakens even where it is not tied. On those cases Qwen3-8B's stated probability no longer beats agreement (0.611 against 0.617): its advantage comes from cases where all samples agree, whereas OLMo-3-7B-Instruct's stated probability stays ahead (0.729 against 0.605). *List size.* The three most frequent of 32 samples can be fewer than three

Model	Valid (%)	Conditions likely + mnm	Recall@5 default	Recall@5 likely part	Severe named (ref. ≥ 0.1) base	Severe named (ref. ≥ 0.1) default	Severe named (ref. ≥ 0.1) mnm	Unwarranted severe default	Unwarranted severe mnm	Acute pulm. edema default	Acute pulm. edema mnm
Qwen3-8B	100	4.1 + 2.3	0.802	0.715	0.318	0.437	0.525	1.06	2.11	0.61	0.60
Qwen2.5-7B-Instruct	99	3.9 + 2.9	0.735	0.640	0.304	0.439	0.503	1.57	2.95	0.63	0.71
Qwen2.5-32B-Instruct	100	4.8 + 2.9	0.840	0.838	0.414	0.472	0.579	0.91	2.27	0.44	0.63
OLMo-3-7B-Instruct	96	3.7 + 2.5	0.485	0.501	0.332	0.317	0.444	1.15	2.60	0.31	0.58
OLMo-3.1-32B-Instruct	100	4.9 + 3.0	0.779	0.790	0.376	0.392	0.545	0.74	2.85	0.41	0.59
Mistral-7B-Instruct	87	3.7 + 2.7	0.613	0.562	0.352	0.329	0.473	0.86	2.29	0.59	0.68

Table 18: Must-not-miss prompt (DDXPlus, T=0.7, 980 cases, 32 samples): the five most likely conditions, then up to three dangerous conditions to rule out. *Default*: the same model with the standard zero-shot prompt; *base*: its base model with the few-shot prompt. Recall@5 of the most-likely part; severe conditions and unwarranted severe conditions (severe, reference < 0.1, per list) over the whole list; acute pulmonary edema: share of lists naming it where the reference gives it any weight.

answers. With $m = \min(3, \text{distinct sampled answers})$, the first m items of one list contain the true pathology about as often as the m most frequent samples for the post-trained models (-0.5 to -5.2 points; significantly lower for Qwen2.5-32B-Instruct and OLMo-3.1-32B-Instruct), while for base models the samples win by 8.5–14.6. The advantage of a post-trained list's top three is that it names three conditions where its samples offer fewer. *Validity.* Counting invalid samples as wrong (as everywhere in the paper) or scoring valid samples only leaves pass@32 unchanged and moves pass@1, mostly for OLMo 3 7B, whose single-answer gain is +7.1 points with invalid samples counted as wrong and -3.5 (n.s.) over valid samples only, weighted to the natural prevalence below. *Prevalence.* The test sample is class-balanced (20 cases per pathology). Reweighted to the pathology frequencies of the full DDXPlus test split (134,529 patients), post-training costs 4.9–31.5 points of pass@32 (all intervals exclude zero, including both 32B pairs), raises recall@5 by 10.3–17.7 points for the Qwen models and OLMo 3.1 32B (-0.3 for OLMo 3 7B) and lowers one list's TV by 0.047–0.069 (OLMo 3 7B +0.031). *Reference ceiling.* The reference's own top five contains the true pathology in 96.9% of cases (97.3% at natural prevalence), against 0.47–0.84 for one model list. *What replaces acute pulmonary edema.* On the 54 cases whose reference gives acute pulmonary edema at least 0.1, the conditions whose share of lists rises most after post-training are mostly other must-not-miss conditions (pulmonary embolism for Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct, +0.29 and +0.52; pneumonia and pulmonary embolism for Qwen2.5-32B-Instruct; unstable angina, panic attack and NSTEMI/STEMI for Qwen3-8B), but atrial fibrillation and influenza for OLMo-3.1-32B-Instruct, whose lists name acute pulmonary edema 0.25 less often.

4.16 A third family: Mistral-7B v0.3

We repeat the comparison with Mistral-7B-v0.3 and Mistral-7B-Instruct-v0.3, a family also studied by Jang et al. [13]. With the zero-shot prompt the base model does not answer: 52% of its samples at T=0.7 and 76% at T=1 are invalid (it continues the document instead), and its valid answers mostly repeat the first listed condition, so its single-call accuracy is 6% and the zero-shot pair is not informative. With the few-shot prompt of §4.10 the base model answers validly (0.4% invalid), and the pair reproduces the other families (Table 22). Post-training raises pass@1 by 14.0 points and lowers pass@32 by 14.5, cuts distinct answers from 5.0 to 1.3 and Cov@32 by 0.138, and moves the samples away from the reference (TV +0.058). Voting gains the base model 5.6 points and the Instruct model 0.8; the three most frequent of 32 samples beat one list by 12.6 points for the base model, while for the Instruct model the list is ahead by 7.0; agreement is the better error signal for the base model (AUROC 0.826 against 0.671) and the stated probability for the Instruct model

(0.747 against 0.613). The stated list improves modestly (TV −0.036 and DDF1 +0.046, surviving clustering and Holm correction over 42 tests; −0.023 with the true pathology removed, which does not), the smallest significant gain of the three families, and 11 of the 17 severe conditions are named less often (3 significantly). Mistral-Instruct also names 0.5 conditions per list that are not among the 49.

4.17 Thinking mode

Qwen3-8B can reason in a thinking block before it answers; the main runs turn this off. With thinking on (zero-shot prompt, T=0.7, up to 8,192 tokens; traces average about 1,500 tokens and all end before the limit), the same weights behave differently (Table 22). Accuracy rises at both ends: pass@1 is 0.671 against 0.559 without thinking, and pass@32 is 0.815, above the non-thinking model (0.607) and level with the base model (0.732 zero-shot, +8.4 [1.7, 15.4] points, not surviving Holm correction over 42 tests; 0.809 with worked examples). The samples are still narrow (1.76 distinct answers against 3.77 for the base model; Cov@32 0.283 against 0.323) but no farther from the reference than the base model's (TV 0.745 against 0.750). The stated list is the closest of any 7–8B model (TV 0.670, against 0.700 without thinking). The shift in error signals reverses: agreement across 31 samples is again the better signal (AUROC 0.732 against 0.638; difference 0.094 [0.027, 0.170]), as for base models, and voting gains 1.5 points (n.s.). The narrowing of the samples and the advantage of the stated probability are therefore properties of answering directly, which a reasoning mode largely undoes, consistent with the chain-of-thought prompt of §4.3.

4.18 Averaging stated lists across calls

Stated lists can be averaged across calls, as samples can be voted. For each case we average the first 5 or 32 valid stated lists, each renormalized over the conditions it names, and score the mean against the reference (Table 23). Averaging helps base models far more than post-trained ones, whose lists vary little between calls (at T=0.7 with the zero-shot prompt, 32 averaged lists name 7.6–13.4 conditions, against 16.6–22.0 for base models). The mean of five base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700, Qwen2.5-7B 0.717 against 0.717, OLMo 3.1 32B 0.691 against 0.701). With five lists on both sides, the post-trained mean is closer for Qwen2.5-7B under both prompts (−0.025 and −0.027) and for Qwen2.5-32B and Qwen3-8B under one prompt each, not significantly closer for OLMo 3.1 32B (zero-shot) and Mistral (few-shot), and farther for OLMo 3 7B and few-shot OLMo 3.1 32B. Averaged base lists also name severe conditions at least as often, and more often for OLMo 3.1 32B, OLMo 3 7B, Mistral and few-shot Qwen2.5-32B (by 0.06–0.14). One post-trained call thus gets about as close to the reference as several base calls; the packing test (Table 24) shows that it is a stabilized version of the base model's view of the patient rather than its literal average, and, as for the open-ended questions of Lake et al. [19], post-training adds little that averaged base lists lack (five post-trained lists beat five base lists in four of eleven settings). Qwen3-8B's thinking mode goes furthest: five of its lists beat five base-model lists by more than any direct-answer model's (TV −0.049) and name more severe conditions (+0.129). For a post-trained model answering directly, averaging five lists instead of reading one improves TV by only 0.005–0.025 at T=0.7 (up to 0.038 at T=1); it names 6–17 points more severe conditions, mostly by naming 1.4 to 3.6 more conditions. Five post-trained lists still cover 0.44–0.50 of the reference mass at T=0.7 (OLMo 3 7B 0.37) with six to eight conditions, about twice what their 32 samples cover (0.22–0.28) and as much as a uniform guesser naming about 24 conditions (0.47), so the coverage deficit of §4.2 is one of the sampled readout.

4.19 Which signal to use: combination and held-out audit

Whether the stated probability or agreement across samples is the better error signal varies across models, so we ask what predicts it and whether a practitioner can tell. We pool 56 model–setting combinations scored on DDXPlus: the base/post-trained pairs at both temperatures and with three prompts, the OLMo stages, the 32B models, the medical models (including three fine-tuned from Llama-3.1-8B, HuatuoGPT-o1-8B, AlphaMed-8B and UltraMedical-8B, which appear only here), our GRPO runs, and the full-distribution prompt. They share weights, prompts and temperatures, so we treat the correlations below as descriptive. The stated probability's advantage over five-sample agreement correlates with the *single-answer rate*, the share of cases whose five samples all give the same top answer (Spearman 0.75), with the stated probability's own AUROC (0.82; the gap contains this term, so part of this is by construction), and negatively with the number of distinct answers (-0.74). The stated probability is the better signal for 19 of the 56. No single-answer threshold separates them: Qwen2.5-7B-Instruct still favors agreement, while OLMo 3 7B's DPO and Instruct models favor their unusually informative stated probability. With five-sample agreement, as in the original audit, combining the signals beats choosing between them; with agreement across all 31 further samples the advantage shrinks (post-trained combinations: agreement 0.742, rank average 0.761, $+0.019$ $[-0.007, 0.040]$; logistic 0.772, $+0.030$ $[0.013, 0.044]$), and for the 17 base combinations agreement alone (0.821) beats the rank average (0.780) and ties the logistic combination (0.827). The rest of this section uses five-sample agreement. Fitting a logistic model of correctness on the stated probability and five-sample agreement with 100 labeled cases, and evaluating on the other 880, gives a mean held-out AUROC of 0.753 for the 39 post-trained combinations (0.765 for all 56); averaging the two signals' ranks without fitting gives 0.746; picking the better single signal on the same 100 cases gives 0.720; always using agreement 0.695; always using the stated probability 0.671. The logistic combination beats the single-signal pick for 35 of the 39 post-trained combinations. This matches the general finding that hybrids of consistency and verbalized confidence beat either [44]. For choosing a single signal, the simplest rule, always using agreement, is right for 66% of the combinations (54% of the 39 post-trained ones) and loses 0.025 AUROC on average (0.032). An audit does better: choose the signal with the higher AUROC on 100 random labeled cases, using five samples per case, and evaluate the choice on the other 880. Over 500 draws per combination it picks the better held-out signal 86% of the time (82% for post-trained models) and loses 0.005 AUROC (0.007); with 50 cases, 82% and 0.010; with 200, 90% and 0.003. It is reliable when the two signals differ by more than 0.05 AUROC (96% of draws, 39 combinations) and close to a coin flip when they differ by less than 0.02 (53%, 5 combinations), where the choice matters least. The audit's main cost is gold labels for 100 cases from the target population, and we validate it only within DDXPlus.

4.20 Medical post-training

Applied after instruction tuning, medical RL does not cost coverage. AlphaMed improves on its parent at every k (Figure 4): $+3.9$ points at k=1 ($[+2.2, +5.5]$; clustered $[-0.4, +8.4]$), widening to $+8.4$ at k=32 ($[+6.1, +10.6]$; clustered $[+2.6, +14.7]$). It reaches at least one correct sample for 47 of 49 conditions, against 39 for its parent. It does not restore breadth, though: distinct answers (1.87 vs. 1.82), entropy (0.28 vs. 0.30), and coverage (0.27 vs. 0.24) stay at the instruction-tuned level, far below the base model. AlphaMed also names off-list conditions more often than its parent, a mild loss of instruction following. By condition, AlphaMed slightly *raises* consideration in every severity tier ($+2.2$ to $+4.3$ points; Table 3). It still loses acute pulmonary edema (0.31 to 0.13), and it introduces an attractor of its own: spontaneous pneumothorax as the top answer in 10.3%

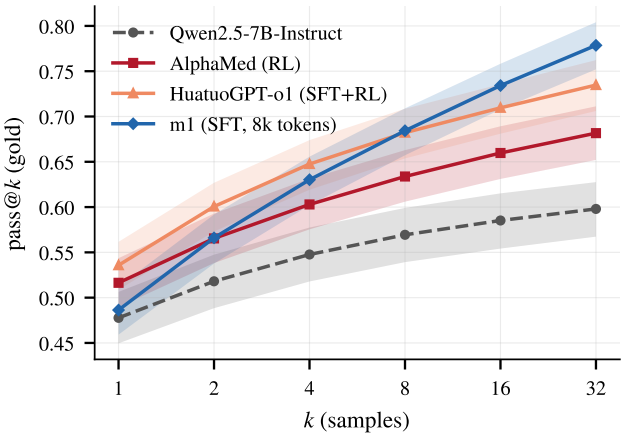


Figure 4: DDXPlus at T=0.7 (980 cases): Qwen2.5-7B-Instruct (dashed) vs. its medical descendants, with 95% case-bootstrap bands. Every descendant matches or beats its parent at every k, and the gaps widen with k: none of the three shows a sharpening tax at the medical stage.

of samples, against 4.3% for its parent and a reference rate of 1.0%. Medical post-training thus operates on an already collapsed distribution; the collapse comes from the general post-training stage. The other two descendants go further (Table 26). HuatuoGPT-o1 (SFT then PPO) and m1 (SFT distillation from a reasoning teacher) both *widen* the differential relative to their parent: 2.49 and 3.29 distinct answers vs. 1.82, and coverage@32 of 0.31 and 0.33 vs. 0.24. Their pass@32 rises by 13.7 [+11.3, +16.0] and 18.1 [+15.4, +20.8] points. m1 gains almost nothing at k=1 (+0.9, interval includes zero), so its improvement lies entirely in breadth. Both reasoning-trained models are also less calibrated (ECE 0.17 and 0.21 vs. 0.11). In these three models, medical post-training on an instruction-tuned model does not add to the collapse, and reasoning distillation partially reverses it.

Medical RL from the base model. Alongside the instruct-initialized AlphaMed-7B, the AlphaMed authors released a companion checkpoint, AlphaMed-7B-base-rl, initialized from Qwen2.5-7B rather than Qwen2.5-7B-Instruct (according to its configuration file). The paper [22] describes only the instruct-initialized models: GRPO with a binary correctness reward on about 19k multiple-choice questions from MedQA and MedMCQA, with a 64-question by 8-sample batch for 300 steps. It does not document the base-initialized checkpoint, so we cannot rule out differences in data or hyperparameters, and we treat it as corroborating rather than controlled evidence. It offers a third-party check on whether RL itself causes the collapse (Table 26, lower block). It does not. RL from the base model improves on the base at every k (+14.3 [+13.1, +15.7] points at k=1, +9.5 [+7.4, +11.6] at k=32). It keeps 3.14 distinct answers, against 1.87 for the instruct-initialized AlphaMed. Against that instruct-model AlphaMed, it is 2.2 [0.4, 3.9] points worse at k=1 and 10.8 [+8.6, +13.2] points better at k=32. Its severe-tier Considered@32 shows no detectable change relative to the base (−0.7 [−3.0, +1.7] points), against −12.4 for instruction tuning and −8.1 for instruction tuning followed by AlphaMed's RL (Table 3; all well below the chance levels of §4.2). RL from a base model also narrows less in OLMo's RL-Zero at T=1 and in our GRPO runs (§5), but not uniformly: RL-Zero at T=0.7 loses about as much as SFT, and within OLMo 3 7B's instruct pipeline the RL stage after SFT and DPO is the largest single narrowing (§4.4).

4.21 Rare disease: overconfidence without accuracy

On 1,401 Phenopacket Store cases covering 749 diseases, every 7–8B model is near the floor: pass@1 is 1.4–2.2% and pass@32 3.7–7.0% for the Qwen-family models, and lower for OLMo 3 7B (2.5–4.5%). Stated confidence is not near the floor. Qwen2.5-7B-Instruct states a mean top-1 probability of 0.51, and its RL-trained medical descendants state 0.67 (HuatuoGPT-o1) and 0.69 (AlphaMed), while their accuracy is about 2%. ECE accordingly rises from 0.49 to 0.64–0.67; at this accuracy ECE mostly restates mean confidence, so we read it as a rise in confidence without a matching rise in accuracy. The RL-trained medical models (AlphaMed; HuatuoGPT-o1 after SFT) thus state much higher confidence where they are least accurate. The effect also appears without instruction tuning: AlphaMed's base-initialized checkpoint is the most overconfident Qwen-family model (mean confidence 0.71, ECE 0.70, accuracy 1.7%). The medical descendants do gain a little accuracy: AlphaMed beats its parent at every k (+0.6 [+0.2, +0.9] points at k=1, +1.0 [+0.1, +2.1] at k=32), HuatuoGPT-o1 at $k \leq 8$, and m1 at $k \geq 2$ (+1.4 [+0.4, +2.5] at k=32). These gains are an order of magnitude smaller than the rise in confidence. m1, trained only with supervised reasoning traces (at its intended 8k-token budget), is the exception on calibration and diversity: it is the most diverse model (15.3 distinct answers per 32 samples, against 11.5 for the parent and 9.2 for HuatuoGPT-o1), and its confidence rises least (0.57).

General post-training reproduces the DDXPlus pattern on Qwen3. Qwen3-8B beats its base at pass@1 (+0.7 [+0.4, +1.1]) and trails it from k=8 on, by −2.2 [−3.1, −1.3] points at pass@32 (a relative loss of 37%), with half the distinct answers (6.4 vs. 13.4). For Qwen2.5 no pass@k difference is significant. Unlike medical RL, general post-training *lowers* stated confidence here (0.61 to 0.46 for Qwen3, 0.61 to 0.51 for Qwen2.5), although base models also give invalid answers more often (8.5% of Qwen3-8B-Base samples). OLMo 3 does not share this: every stage of its pipeline states a mean confidence of 0.73–0.84 at about 1% accuracy (pass@32 2.5–4.5%), and its SFT checkpoint is the most overconfident model we evaluate (confidence 0.84, ECE 0.83). On the 500-case subset, our 8B GRPO runs (§5) are indistinguishable from their base (all pass@k differences within ±0.6 points).

Accuracy depends strongly on when the source case report was published (Table 27). For every model, pass@32 is 5–10% on cases published up to 2022 (n=901), at most 1.7% on 2023–24 cases (n=348), and 2.6–4.6% on 2025+ cases (n=152). The pattern is not monotone, since 2025+ cases score above 2023–24 cases, so case mix also matters; the gap between older and recent cases is consistent with, but does not establish, memorization of older case reports.

4.22 Published case reports

MedCaseReasoning presentations stop before the authors' differential, so these cases test open-ended diagnosis from real presentations (prompts and grading in Appendix 8). We sampled the four zero-shot Qwen and OLMo pairs; OLMo 3.1 32B was left out for cost and Mistral because its base model needs worked examples, which this dataset cannot supply without reference lists. The two judges agree on 97.6% of 391,987 distinct names (κ 0.77); Phi-4 accepts more names (6.3% against 4.5%), and the strict grading leaves every direction unchanged, though Qwen2.5-32B's pass@32 loss and OLMo 3 7B's recall gain lose significance. The authors' alternatives were extracted by a language model from the reasoning notes; some are broad ("bacterial", "viral"), which only makes them easier to match. Against this clinician-written reference the stated/sampled split replicates (Table 28): post-trained samples cover 3–14 points fewer of the authors' alternatives, one list covers them as before, and for Qwen3-8B and Qwen2.5-32B one list covers more of them than 32 samples, whereas base models' samples cover more than their lists (by 0.5–10.0 points, n.s. for Qwen2.5-

32B). The consequences differ from DDXPlus: majority voting still adds 2.5–11.8 points for the post-trained models (7.4–7.9 for the base models), agreement is the better error signal for every model, and post-training lowers its AUROC only for Qwen3-8B (0.853 → 0.788) while raising it for the others. OLMo-3-7B-Instruct gives more distinct answers than its base model (22.5 against 13.1 among valid samples), whose answers are valid only 67% of the time. Accuracy is not lower for cases published from 2023 (e.g. Qwen2.5-32B-Instruct pass@32 0.49 up to 2022 and 0.52 from 2023), so these results show no sign of memorized reports.

4.23 Stated and sampled readouts on real rare-disease cases

Phenopacket Store cases are real, but they have no reference differential and accuracy is near the floor, so we can test only the consequences of §4.7, and we count an answer as correct if it is the diagnosis or a closely related Mondo class (exact matches give only 8–31 correct single-call answers per model). Top answers are grouped by Mondo class for voting and agreement (Table 29). The DDXPlus shifts reappear, but the top-three comparison does not flip as it does on DDXPlus. Majority voting over 32 samples helps the base models by 3.1–4.6 points but the final post-trained models by only 0.4–1.5 (Qwen3-8B +0.4, n.s.). Post-training weakens agreement as an error signal (Qwen3-8B 0.830 → 0.763, Qwen2.5-7B 0.845 → 0.745, OLMo 3 7B 0.814 → 0.726) and strengthens the stated probability (0.567 → 0.707, 0.598 → 0.717, 0.464 → 0.652). Agreement beats the stated probability by 0.25–0.35 AUROC for the base models and by only 0.03–0.07 for the post-trained ones, significantly only for Qwen3-8B (-0.056 $[-0.120, -0.002]$). On these open-ended cases the three most frequent of 32 sampled answers still contain a correct diagnosis more often than one list's top three, but the gap shrinks from 4.6–7.0 points for the base models to 2.7–3.6 for Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct and to a tie for Qwen3-8B ($+0.4$ $[-0.8, +1.8]$). The medical models behave like their post-trained parent, with agreement the better signal for all of them.

5 Training without collapsing the samples

Related rewards. Closely related GRPO rewards computed from group answer counts have appeared since these experiments: a witness advantage that trains models to sample the distributions they state [1], and a batch bonus that penalizes repeated answers [36]. Our group reward `gm` additionally needs a reference differential at training time, which real clinical data rarely provide; we do not compare against them, and we treat the results below as existence results.

The stated list survives post-training, but sampled diversity still matters: self-consistency and other sampling-based methods depend on it (§4.7), and so does reinforcement learning itself, since GRPO learns nothing from a group whose samples agree. Can reinforcement learning raise diagnostic accuracy without collapsing the sampled differential? We fine-tune Qwen3-1.7B with GRPO on DDXPlus training patients (9,800 cases, 200 per condition; disjoint from the test split) and evaluate exactly as in §3.

Setup. We train LoRA adapters (r=32) with GRPO for 200 steps. Each step uses 16 prompts and 16 sampled completions per prompt, generated with vLLM on the same GPU. A first sweep with a common aggressive configuration (learning rate 2×10^{-5}, no KL penalty, 8 samples per prompt, per-group reward standardization) was unstable. In three of seven runs, the policy collapsed to an input-independent answer: one condition for 88–100% of all patients, with completions shrinking to about 25 tokens. The starting policy is nearly deterministic (token entropy ≈0.06), so most GRPO groups have zero reward variance, and noisy updates drive the collapse. We therefore use

learning rate 5×10^{-6}, KL coefficient 0.04, no reward standardization (as in Puri et al. [32]), and 16 samples per prompt. A monitor tracks the number of distinct top-1 answers per case within each batch. All runs in this configuration were stable.

Rewards. Every recipe includes a format reward: the answer must parse into a list of distinct conditions with valid probabilities. `acc` is standard RLVR: 1 if the top-1 answer is the true pathology. `dist` adds two per-sample distributional terms to `acc`: a Brier score of the stated distribution against the truth, and the overlap of the stated distribution with the reference differential. Both are computed from a *single* completion. They cannot see that the policy gives the same answer in every sample, which is exactly the collapse of §4.2. `gm` adds to `acc` a *group-level* reward over the GRPO group of n samples for a case. If sample i's top-1 answer is a, and k_a samples in the group give a,

$$r_i^{\mathrm{gm}} = \mathrm{clip}\big(n\, p_{\mathrm{ref}}(a) - (k_a - 1),\ 0,\ 1\big), \tag{1}$$

where p_{ref} is the reference differential. This is sample i's marginal contribution to $n\sum_c \min(q_c, p_{\mathrm{ref}}(c))$, the overlap between the group's empirical answer distribution q and the reference. An answer earns credit while its condition is under-represented relative to its reference mass, and none once it is over-represented. In contrast to multi-answer rewards that score a set inside one completion [32], `gm` acts on the distribution of *sampled* answers, the quantity that collapses. Because the group reward fires only when a group contains dissenting samples, we also test a boosted variant, `gm`$^+$: `gm` weight 3 and training temperature 1.2. As baselines we adapt the multi-answer rewards of Puri et al. [32] to our prompt. Multi-RLVR scores the share of listed conditions that belong to the reference differential, treated as a set of acceptable answers; Multi-RLCR adds their set-level Brier term on the per-answer probabilities. Both zero the reward for duplicate or invalid answers, as in the original, and use the same format reward, optimizer and settings as our recipes.

Results. Table 31 reports means over two seeds; intervals bootstrap cases and do not include seed-to-seed variation. All our recipes raise top-1 accuracy by 8–10 points; the multi-answer baselines barely move it (below). The per-sample recipes (`acc`, `dist`) slightly *reduce* diversity relative to the starting model, the sharpening documented in §4.1. The group reward removes this. `gm` matches the starting model's diversity, and `gm`$^+$ exceeds it while also being the most accurate recipe. Against `acc`, with paired bootstraps over cases averaged over seeds, `gm`$^+$ improves:

- pass@1 by $+1.6$ $[+1.0, +2.3]$ and pass@32 by $+1.7$ $[+1.0, +2.6]$ points;
- distinct answers by $+0.10$ $[+0.07, +0.13]$ and entropy by $+0.034$ $[+0.022, +0.047]$;
- coverage@32 by $+1.5$ $[+1.0, +2.1]$ points, and TV by -0.014 $[-0.019, -0.010]$.

Plain `gm` improves on `acc` in the same direction with smaller margins (e.g. pass@1 $+1.0$ $[+0.6, +1.6]$, entropy $+0.013$ $[+0.005, +0.021]$). Diversity increases monotonically with group-reward pressure, with no accuracy cost. The set-membership baselines of Puri et al. [32] train stably (their reward rises from 0.27–0.31 to 0.32–0.35) but barely change sampled behavior. pass@1 rises by under 1 point over the untrained model, and diversity is unchanged. Against them, `gm`$^+$ is better on every metric, by $+8.9$ $[+7.4, +10.5]$ points at pass@1, $+0.07$ $[+0.03, +0.11]$ distinct answers and $+3.0$ $[+2.3, +3.8]$ points of coverage@32 (vs. Multi-RLVR; Multi-RLCR is similar). What they do improve is the differential written *inside* one completion: the reference mass covered by a single listed differential rises to 0.317 (Multi-RLVR), against 0.300 for `acc` and 0.309 for `gm`$^+$. The two objectives are complementary: a set reward shapes the list a model writes, while the group

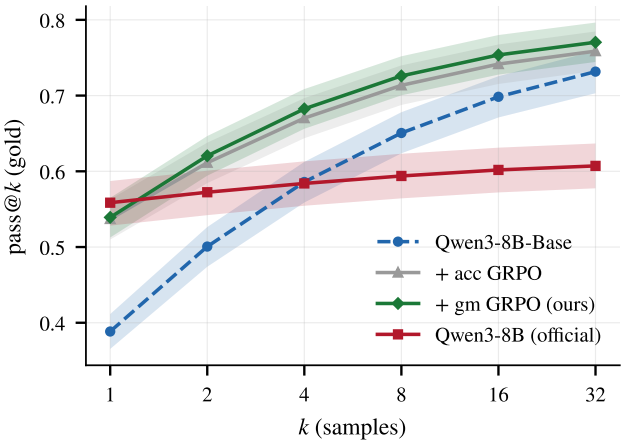


Figure 5: pass@k on DDXPlus (T=0.7, 980 cases) from the same pretrained weights: Qwen3-8B-Base, our GRPO runs (pooled over two seeds), and the official post-trained Qwen3-8B, with 95% case-bootstrap bands. Our RL lifts the whole curve; official post-training trades the right end for the left.

reward shapes which answer it commits to across samples, which is what collapses under post-training. The absolute effects are modest because the starting policy is already collapsed: there is little exploration for any reward to act on. None of the recipes changes rare-disease accuracy or confidence on Phenopacket Store, where a 1.7B model is at the floor; the same holds at 8B (§4.21).

Starting from a diverse base model. The 1.7B experiment starts from an already collapsed instruct model. We therefore repeat the comparison from Qwen3-8B-Base, the diverse base model of §4.1, with the same recipe and `acc` vs. `gm`. We compare against Qwen's own post-trained Qwen3-8B (Table 30; means of two seeds). Both RL recipes reach nearly the official model's top-1 accuracy (0.537 and 0.539 vs. 0.558; `gm`: -1.9 $[-3.9, +0.1]$ points). They keep more than twice its diversity (2.7–2.9 vs. 1.27 distinct answers) and beat it by 15–16 points at pass@32 (`gm`: +16.3 $[+14.0, +18.8]$). In this setting, then, the collapse produced by general post-training is not required for its top-1 accuracy (our runs train on in-distribution DDXPlus data, so this is an existence result, not a like-for-like comparison of recipes). Both runs still narrow the base model somewhat, and the group reward narrows it less. At equal pass@1 (+0.2, n.s.), `gm` beats `acc` on distinct answers $+0.14$ $[+0.11, +0.17]$, entropy $+0.035$ $[+0.029, +0.041]$, coverage@32 $+1.1$ $[+0.8, +1.5]$ points, and pass@32 $+1.2$ $[+0.5, +1.9]$. Each seed reproduces the ordering. The severe tier tells the same story (Table 3): official post-training removes 9.8 points of can't-miss consideration, `acc` 2.8, and `gm` 1.5 (not significant). Figure 5 shows the pass@k curves.

6 Claims and the evidence for them

Table 32 lists each claim with its primary evidence, the checks under which it holds and those under which it weakens or fails; Table 33 gives the same information control by control. Primary evidence comes from the 112 headline tests (§7); everything else is a secondary check.

Pair	Union set (7 conditions) base	post	post − base	sig. lost / gained	Harm-based post − base	Chest pain post − base
Qwen3-8B, T=0.7	0.494	0.584	+0.089	0 / 2	+0.053	+0.115
Qwen3-8B, T=1	0.425	0.585	+0.160	0 / 4	+0.135	+0.167
Qwen2.5-7B, T=0.7	0.315	0.536	+0.221	0 / 4	+0.267	+0.149
Qwen2.5-7B, T=1	0.303	0.539	+0.236	0 / 4	+0.287	+0.180
Qwen2.5-32B, T=0.7	0.432	0.595	+0.162	0 / 4	+0.164	+0.104
OLMo 3 7B, T=0.7	0.268	0.413	+0.144	2 / 3	+0.133	+0.101
OLMo 3 7B, T=1	0.275	0.412	+0.136	3 / 3	+0.135	+0.102
OLMo 3.1 32B, T=0.7	0.462	0.459	−0.003†	3 / 0	−0.033†	−0.020†
Qwen3-8B, few-shot	0.299	0.491	+0.191	0 / 3	+0.183	+0.093
Qwen2.5-7B, few-shot	0.239	0.455	+0.216	2 / 3	+0.249	+0.152
Qwen2.5-32B, few-shot	0.361	0.542	+0.180	1 / 3	+0.210	+0.096
OLMo 3 7B, few-shot	0.335	0.409	+0.074	2 / 3	−0.015†	+0.036†
OLMo 3.1 32B, few-shot	0.412	0.477	+0.065†	2 / 1	+0.080	−0.017†
Mistral-7B, few-shot	0.401	0.441	+0.040†	4 / 1	+0.024†	−0.071

Post-trained model	Warranted must-not-miss conditions named (union set) standard	two lists	must-not-miss	− standard	Unwarranted per list standard	two lists	must-not-miss
Qwen3-8B	0.584	0.612	0.775	+0.192	1.08	1.20	2.10
Qwen2.5-7B-Instruct	0.536	0.600	0.714	+0.178	0.94	1.14	2.33
Qwen2.5-32B-Instruct	0.595	0.648	0.814	+0.219	1.00	1.21	2.47
OLMo-3-7B-Instruct	0.413	0.479	0.564	+0.151	0.93	1.17	1.60
OLMo-3.1-32B-Instruct	0.459	0.527	0.715	+0.256	0.58	0.77	1.77
Mistral-7B-Instruct	0.358	0.417	0.615	+0.257	0.69	0.87	1.65

Table 19: Must-not-miss conditions defined outside DDXPlus (DDXPlus, 980 cases). *Harm-based*: the DDXPlus conditions among the 15 diseases behind about half of serious misdiagnosis-related harms in the USA [29] (NSTEMI/STEMI, pulmonary embolism, pneumonia, pulmonary neoplasm). *Chest pain*: the life-threatening causes of acute chest pain in the 2021 AHA/ACC guideline [10] (NSTEMI/STEMI, unstable angina, pulmonary embolism, Boerhaave, spontaneous pneumothorax). *Union*: either set. Top: share of the set's conditions with reference probability ≥ 0.1 that one list names, and the number of the union's conditions named significantly less / more often (Benjamini–Hochberg over the 7 conditions). Bottom: the must-not-miss prompt (§4.12) scored against the union set; *two lists*: the union of two standard lists, a length-matched comparison; unwarranted: the set's conditions named that the reference weights below 0.1. †: the condition-clustered 95% interval includes zero.

Pair	Zero-shot shuffled (base / post)	Zero-shot raw	Zero-shot corrected	Worked examples (reference lists) shuffled (base / post)	Worked examples (reference lists) raw	Worked examples (reference lists) corrected	Worked examples (another model's lists) shuffled (base / post)	Worked examples (another model's lists) raw	Worked examples (another model's lists) corrected
Qwen3-8B	0.063 / 0.014	−12.4	−7.5	0.104 / 0.012	−17.1	−7.9	0.072 / 0.010	−14.9	−8.7
Qwen2.5-7B	0.088 / 0.026	−9.8	−3.6	0.087 / 0.022	−22.9	−16.4	0.080 / 0.022	−20.1	−14.3
Qwen2.5-32B	0.063 / 0.018	−4.9	−0.5	0.056 / 0.015	−15.3	−11.2	0.049 / 0.014	−14.5	−10.9
OLMo 3 7B	0.088 / 0.039	−30.1	−25.2	0.163 / 0.041	−30.0	−17.8	0.191 / 0.034	−29.9	−14.3
OLMo 3.1 32B	0.072 / 0.029	−4.3	+0.1	0.083 / 0.020	−19.1	−12.7	0.095 / 0.020	−17.6	−10.0
Mistral-7B	—	—	—	0.091 / 0.017	−14.1	−6.7	0.097 / 0.017	−12.1	−4.1

Table 20: Chance-corrected pass@32 (DDXPlus, T=0.7, 980 cases). *Shuffled*: pass@32 when each case's samples are scored against the true pathology of a random case with a different pathology (mean of 20 permutations), the part of pass@32 that does not depend on the patient. *Raw*: post-trained minus base pass@32, in points; *corrected*: the same after subtracting each model's shuffled value. Mistral's base model does not answer the zero-shot prompt (§4.16). This check takes a sample's top answer to be its first listed name that maps to one of the 49 conditions, so raw values can differ from Tables 2 and 13 by up to 1.8 points.

Post-trained model of	TV to own base mean	Closest other base model	Closest other base model own − it	Closest other post-trained model	Closest other post-trained model own − it	Own − mean of other post-trained models
Qwen3-8B	0.368	OLMo-3-1125-32B	−0.058	Qwen2.5-32B-Instruct	+0.000†	−0.067
Qwen2.5-7B	0.464	Qwen3-8B-Base	+0.014†	Qwen2.5-32B-Instruct	+0.028	−0.041
Qwen2.5-32B	0.360	Qwen3-8B-Base	−0.036	OLMo-3.1-32B-Instruct	−0.018†	−0.090
OLMo 3 7B	0.628	Qwen3-8B-Base	+0.031	Qwen3-8B	+0.041	+0.002†
OLMo 3.1 32B	0.386	Qwen2.5-32B	−0.051	Qwen2.5-32B-Instruct	−0.006†	−0.079

Table 21: A stricter packing test (DDXPlus, zero-shot, T=0.7, 980 cases). For each post-trained model, the TV between its lists and the mean list of each other final model (five base, five post-trained) for the same patient. Negative differences: the post-trained list is closer to its own base model's view than to the comparison. †: the condition-clustered 95% interval includes zero.

Model	Invalid (%)	pass@1	pass@32	Distinct	Cov@32	TV, one list all	TV, one list w/o true	DDF1	TV samples	Vote@32 gain	Top-3 list − samples	AUROC stated	AUROC agree@5
Mistral-7B-v0.3 (base), few-shot	0.4	0.428	**0.791**	**5.02**	**0.355**	0.777	0.875	0.250	**0.741**	+5.6	−12.6	0.671	0.826
Mistral-7B-Instruct-v0.3, few-shot	0.3	**0.568**	0.646	1.34	0.217	**0.741**	**0.851**	**0.296**	0.799	+0.8	+7.0	0.747	0.613
Qwen3-8B-Base	11.3	0.388	0.732	**3.77**	**0.323**	0.743	0.833	0.276	0.750	+3.4	−7.3	0.595	0.772
Qwen3-8B, non-thinking	0.0	0.559	0.607	1.27	0.217	0.700	0.815	0.318	0.793	+0.3	+13.7	0.734	0.613
Qwen3-8B, thinking	0.0	**0.671**	**0.815**	1.76	0.283	**0.670**	**0.799**	**0.349**	**0.745**	+1.5	+3.0	0.638	0.693

Table 22: A third family and a reasoning mode on DDXPlus (T=0.7, 980 cases, 32 samples). Top: Mistral-7B v0.3 with the few-shot prompt of §4.10. Bottom: Qwen3-8B-Base and Qwen3-8B with thinking off (the main runs) and on, with the zero-shot prompt. Columns as in Table 12; Distinct: distinct top answers per 32 samples. Bold: best within each block.

Pair	TV, base model			TV, post-trained model			Post − base, 5 lists each	
	1 list	mean of 5	mean of 32	1 list	mean of 5	mean of 32	TV	Severe named
Zero-shot prompt								
Qwen3-8B	0.743	0.699	0.691	0.700	0.695	0.692	$-0.004^{\dagger}$	$+0.030^{\dagger}$
Qwen2.5-7B	0.779	0.717	0.706	0.717	0.692	0.685	−0.025	$+0.023^{\dagger}$
Qwen2.5-32B	0.726	0.681	0.674	0.674	0.658	0.654	−0.024	$-0.037^{\dagger}$
OLMo 3 7B	0.794	0.730	0.717	0.823	0.805	0.801	+0.074	−0.076
OLMo 3.1 32B	0.746	0.691	0.678	0.701	0.677	0.673	$-0.014^{\dagger}$	−0.088
Few-shot prompt								
Qwen3-8B	0.768	0.704	0.689	0.682	0.673	0.672	−0.030	$-0.006^{\dagger}$
Qwen2.5-7B	0.784	0.712	0.694	0.702	0.686	0.683	−0.027	$+0.012^{\dagger}$
Qwen2.5-32B	0.712	0.655	0.641	0.654	0.640	0.638	$-0.015^{\dagger}$	−0.094
OLMo 3 7B	0.787	0.703	0.676	0.808	0.786	0.778	+0.083	−0.141
OLMo 3.1 32B	0.719	0.651	0.630	0.697	0.682	0.680	+0.031	−0.060
Mistral-7B v0.3	0.777	0.708	0.691	0.741	0.723	0.720	$+0.015^{\dagger}$	−0.118
Zero-shot prompt, thinking on								
Qwen3-8B, thinking	0.743	0.699	0.691	0.670	0.650	0.647	−0.049	+0.129

Table 23: Averaging stated lists across calls (DDXPlus, T=0.7, 980 cases). TV to the reference differential of one stated list (per-case mean over valid lists, as in Table 7) and of the mean of the first 5 or 32 valid lists, for the base and the final post-trained model. Last two columns: post-trained minus base with five averaged lists each, for TV and for the share of severe conditions with reference probability ≥ 0.1 that are named. †: the condition-clustered 95% interval includes zero.

Pair	TV of a post-trained list to			Within-model benchmark		Differences	
	own base mean	same-diagnosis means	reference	base list	post list	own − same dx	own − base list
Zero-shot prompt							
Qwen3-8B, T=0.7	0.368	0.444	0.701	0.372	0.109	−0.076	$-0.004^{\dagger}$
Qwen3-8B, T=1	0.424	0.486	0.701	0.551	0.153	−0.062	−0.128
Qwen2.5-7B, T=0.7	0.464	0.519	0.717	0.470	0.248	−0.055	$-0.006^{\dagger}$
Qwen2.5-7B, T=1	0.532	0.575	0.717	0.650	0.315	−0.043	−0.119
Qwen2.5-32B, T=0.7	0.360	0.431	0.674	0.375	0.189	−0.071	$-0.015^{\dagger}$
OLMo 3 7B, T=0.7	0.628	0.658	0.823	0.493	0.238	−0.031	+0.135
OLMo 3 7B, T=1	0.651	0.679	0.818	0.691	0.326	−0.028	−0.040
OLMo 3.1 32B, T=0.7	0.386	0.452	0.700	0.432	0.227	−0.066	−0.046
Few-shot prompt							
Qwen3-8B	0.469	0.518	0.681	0.471	0.102	−0.049	$-0.003^{\dagger}$
Qwen2.5-7B	0.559	0.596	0.702	0.515	0.171	−0.037	+0.044
Qwen2.5-32B	0.465	0.511	0.654	0.441	0.154	−0.045	+0.024
OLMo 3 7B	0.669	0.710	0.808	0.594	0.259	−0.040	+0.075
OLMo 3.1 32B	0.482	0.532	0.697	0.469	0.150	−0.050	$+0.014^{\dagger}$
Mistral-7B	0.495	0.546	0.740	0.495	0.181	−0.051	$-0.000^{\dagger}$
Thinking on							
Qwen3-8B, thinking	0.412	0.468	0.669	0.372	0.189	−0.057	+0.040

Table 24: Does a post-trained list reproduce its base model's view of the patient? (DDXPlus, 980 cases.) TV between a post-trained list and the mean of the base model's valid lists for the same case (own base mean), the base model's means for the other cases with the same true pathology, and the reference. Within-model benchmarks: a base list against the mean of the base model's other lists for the case, and a post-trained list against the mean of the post-trained model's other lists. †: the condition-clustered 95% interval includes zero.

Model (T=0.7)	Stated p	Agree@5	Agree@32
Qwen3-8B-Base	+0.042	+0.005	+0.024
Qwen3-8B	+0.116	+0.056	+0.098
Qwen2.5-7B	+0.044	+0.025	+0.038
Qwen2.5-7B-Instruct	+0.070	+0.054	+0.070
Qwen2.5-32B	+0.064	−0.026	−0.038
Qwen2.5-32B-Instruct	−0.043	−0.060	−0.109
OLMo-3-1025-7B	+0.083	−0.048	−0.052
OLMo-3-7B-Instruct	−0.067	−0.073	−0.049
OLMo-3-1125-32B	−0.005	+0.020	+0.011
OLMo-3.1-32B-Instruct	+0.063	+0.014	−0.023

Table 25: Spearman correlation between each uncertainty signal (lower confidence or agreement) and the entropy of the case's reference differential (DDXPlus, 980 cases). Neither readout tracks how ambiguous a case is.

Model	Post-training	pass@1	pass@32	Distinct	Entropy	Cov@32	ECE
Qwen2.5-7B-Instruct	(parent)	0.478	0.598	1.82	0.30	0.239	**0.106**
AlphaMed-7B	rule-based RL	0.516	0.682	1.87	0.28	0.266	0.121
HuatuoGPT-o1-7B	SFT + PPO	**0.536**	0.735	2.49	0.44	0.306	0.168
m1-7B-23K	SFT distillation	0.486	0.779	3.29	0.62	0.328	0.206
Qwen2.5-7B (base)	(parent)	0.352	0.695	**4.91**	**1.00**	**0.335**	0.291
AlphaMed-7B (base-RL)	rule-based RL	0.495	**0.790**	3.14	0.62	0.320	0.202

Table 26: Medical models on DDXPlus (980 cases, 32 samples, T=0.7). Upper block: descendants of Qwen2.5-7B-Instruct; none narrows the differential further, and the reasoning-trained models widen it. Lower block: AlphaMed's companion checkpoint trained from the base model, which keeps most of the base model's diversity and has the best zero-shot pass@32 of any 7B model.

	≤2022 (n=901)		2023–24 (n=348)		2025+ (n=152)	
Model	pass@1	pass@32	pass@1	pass@32	pass@1	pass@32
Qwen2.5-7B (base)	2.0	8.2	0.0	0.3	0.6	3.3
Qwen2.5-7B-Instruct	2.3	7.7	0.0	0.9	0.8	3.9
AlphaMed-7B	3.0	9.0	0.3	1.4	1.4	3.9
AlphaMed-7B (base-RL)	2.4	8.8	0.1	1.1	1.4	4.6
HuatuoGPT-o1-7B	3.1	8.4	0.3	1.4	1.3	3.9
m1-7B	2.6	9.5	0.3	1.7	1.0	3.9
Qwen3-8B-Base	2.1	8.3	0.1	0.6	1.1	3.9
Qwen3-8B	3.1	5.2	0.1	0.3	2.0	2.6

Table 27: Phenopacket Store accuracy (%, exact diagnosis) by publication year of the source case report (1,401 cases, 32 samples, T=0.7). Every model is far more accurate on cases published up to 2022 than on 2023–24 cases; 2025+ cases fall in between.

Pair	Model	Valid	pass@1	pass@32	Distinct	Recall@5 one list	Authors' alternatives one list	Authors' alternatives 32 samples	Vote@32 − single	AUROC, single-call answer correct stated p	AUROC, single-call answer correct agree@32	AUROC, single-call answer correct stated − agree
Qwen3-8B	base	0.878	0.139	0.433	11.95	0.301	0.303	0.326	+7.4	0.558	0.853	−0.294
	post-trained	1.000	0.180	0.308	6.05	0.352	0.296	0.186	+2.5	0.678	0.788	−0.111
Qwen2.5-7B	base	0.986	0.134	0.438	14.61	0.263	0.290	0.378	+7.9	0.565	0.839	−0.273
	post-trained	1.000	0.154	0.407	12.55	0.293	0.285	0.289	+5.7	0.623	0.849	−0.226
Qwen2.5-32B	base	0.985	0.206	0.530	12.19	0.371	0.322	0.326	+7.5	0.666	0.870	−0.204
	post-trained	1.000	0.251	0.495	9.80	0.429	0.320	0.245	+4.3	0.650	0.893	−0.243
OLMo 3 7B	base	0.669	0.073	0.392	13.08	0.222	0.250	0.350	+7.7	0.462	0.832	−0.371
	post-trained	0.994	0.120	0.370	22.47	0.240	0.237	0.319	+11.8	0.675	0.960	−0.284

Pair	Post-trained − base (primary grading) pass@1	pass@32	Distinct	recall@5	alt., list	alt., samples	List − samples (post, alt.)	Vote − single (post)	Strict grading pass@32	recall@5
Qwen3-8B	+4.1	−12.5	−5.90	+5.1	−0.7†	−14.0	+11.0	+2.5	−12.2	+5.0
Qwen2.5-7B	+2.0	−3.1†	−2.06	+3.0	−0.5†	−8.9	−0.4†	+5.7	−3.2†	+2.8
Qwen2.5-32B	+4.4	−3.5	−2.39	+5.8	−0.2†	−8.1	+7.5	+4.3	−3.0†	+5.7
OLMo 3 7B	+4.7	−2.2†	+9.39	+1.8	−1.3	−3.1	−8.2	+11.8	−1.9†	+1.8†

Table 28: Published case reports (MedCaseReasoning test split, 897 PMC case reports, zero-shot, 32 samples, T=0.7). A name is correct if the primary judge (Mistral-Small-24B-Instruct) says it names the report's final diagnosis; *strict*: only if Phi-4 agrees as well (the judges agree on 97.6% of 391,987 judged names, Cohen's κ 0.77). *Authors' alternatives*: share of the diagnoses other than the final one that the report's authors considered (extracted from its reasoning notes; 893 cases with at least one, 3.5 on average) that one list names, or that any of the 32 sampled top answers names; *list − samples*: the difference within the post-trained model. Invalid samples count as wrong for pass@k; the list and agreement columns use valid samples. Distinct: distinct top answers among valid samples, with correct answers merged and others compared as normalized text. Differences in points. †: not significant (paired case bootstrap, 10,000 resamples, Holm over the 36 pair tests; for the AUROC and list − samples differences, the 95% interval includes zero).

Model	Single-call correct	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct Stated p	AUROC, single-call answer correct Agree@5	AUROC, single-call answer correct Agree@32	Correct answers
Qwen3-8B-Base	0.074	+4.6	−7.0	0.567	0.768	**0.830**	104
Qwen3-8B	0.080	+0.4	+0.4	0.707	0.726	**0.763**	112
Qwen2.5-7B	0.062	+3.8	−5.4	0.598	0.788	**0.845**	87
Qwen2.5-7B-Instruct	0.056	+1.5	−3.6	0.717	0.661	**0.745**	78
OLMo-3-1025-7B	0.060	+3.1	−4.6	0.464	0.765	**0.814**	84
SFT	0.035	+1.1	−3.6	0.628	0.730	**0.814**	49
SFT+DPO	0.026	+1.2	−3.3	0.651	0.673	**0.725**	36
SFT+DPO+RLVR (Instruct)	0.035	+0.9	−2.7	0.652	0.644	**0.726**	49
RL-Zero (from base)	0.065	+2.3	−3.1	0.583	0.791	**0.827**	91
AlphaMed-7B	0.071	+1.6	−3.0	0.586	0.755	**0.797**	100
HuatuoGPT-o1-7B	0.093	+2.6	−5.7	0.653	0.708	**0.756**	130
m1-7B	0.084	+2.9	−7.9	0.650	0.747	**0.800**	117
AlphaMed-7B (base-RL)	0.054	+2.5	−4.2	0.672	0.733	**0.789**	75

Table 29: Aggregation and uncertainty on Phenopacket Store (1,401 real rare-disease cases, 32 samples, T=0.7). An answer counts as correct if it is the diagnosis or a closely related Mondo class. Columns as in Table 8; Correct answers: number of correct single-call answers, the positives for the AUROCs. Bold: best signal.

Model	pass@1	pass@32	Distinct	Entropy	Cov@32	TV↓
Qwen3-8B-Base (start)	0.389	0.732	**3.77**	**0.78**	**0.323**	0.749
+ `acc` GRPO (ours)	0.537	0.759	2.72	0.52	0.297	0.750
+ `gm` GRPO (ours)	0.539	**0.770**	2.86	0.56	0.308	**0.742**
Qwen3-8B (official post-training)	**0.558**	0.607	1.27	0.11	0.217	0.793

Table 30: GRPO from Qwen3-8B-Base vs. the official post-trained Qwen3-8B on DDXPlus (980 cases, 32 samples, T=0.7); GRPO rows are means of two seeds. Our RL nearly matches the official model's top-1 accuracy while keeping more than twice its diversity; the group reward preserves more than `acc`. The base row comes from this experiment's own sampling run, so it differs slightly from Table 2.

Recipe	pass@1	pass@32	Distinct	Entropy	Cov@32	TV↓
Qwen3-1.7B (untrained)	0.378	0.435	1.41	0.151	0.162	0.850
`acc`	0.458	0.499	1.36	0.142	0.181	0.829
`dist`	0.458	0.502	1.34	0.131	0.181	0.829
MULTI-RLVR [32]	0.385	0.443	1.39	0.153	0.166	0.845
MULTI-RLCR [32]	0.383	0.441	1.39	0.150	0.165	0.847
`gm`	0.469	0.510	1.40	0.155	0.186	0.825
$\texttt{gm}^{+}$	**0.474**	**0.516**	**1.46**	**0.176**	**0.196**	**0.815**

Table 31: GRPO on Qwen3-1.7B, evaluated on DDXPlus test (980 cases, 32 samples, T=0.7); means of two seeds. $\texttt{gm}^{+}$: group-reward weight 3, training temperature 1.2. Per-sample rewards slightly narrow the model; set-membership rewards barely move it; the group reward preserves, then widens, its differential while improving accuracy.

Claim	Primary evidence	Holds under	Weaker or fails under
(a) A post-trained list stays close to the base model's view of each patient and varies little between calls (§4.6)	A post-trained list is closer to its own base model's mean list for the same patient than to the means for other patients with the same diagnosis, in all 15 settings (TV −0.028 to −0.076); post-trained lists vary 0.10–0.33 between calls, base lists 0.37–0.69 (Table 24)	worked examples; thinking mode; against other families' base models (−0.03 to −0.14); for three of five pairs, against every other base model (Table 21)	about as close to the nearest other post-trained model; Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct not closest to their own base
(b) Sampled answers collapse (§4.2)	pass@32 −9.7 to −14.6 points for the 7–8B Qwen pairs and −28.3 for OLMo 3 7B zero-shot; 1.3–2.4 distinct answers against 3.8–4.9 (Table 2)	worked examples of either kind (13–30 points in all six pairs); Mistral; OLMo stages; 32B with worked examples	32B zero-shot (n.s.); Qwen2.5-7B-Instruct at T=1.3; thinking mode; RL-Zero at T=1
(c) Post-training's first-answer gain is mostly prompt format and sampling noise (§4.2)	With shared worked examples the sampled pass@1 gain is 1.6–5.2 points in four of six pairs; decoded greedily, post-training raises top-1 significantly in at most two of six pairs under either matched prompt (Table 16)	both matched pairings; both 32B pairs	Qwen3-8B (+10.7 and +13.7); Mistral with shared examples (+10.8)
(d) One stated list improves robustly only for the smaller Qwen models (§4.6)	Recall@5 +12 to +24 points, TV −0.043 to −0.088, conditional TV −0.025 to −0.078 (Table 7, Table 34)	complete lists; correct and wrong top answers; true pathology removed; worked examples for Qwen3-8B and Qwen2.5-7B; thinking mode	OLMo 3 7B (no gain); OLMo 3.1 32B and Qwen2.5-32B recall with worked examples; five averaged base lists match one post-trained list
(e) Severe conditions are named more often on average, but specific ones are lost (§4.6)	Named more often in every zero-shot setting (+2.2 to +13.8 points, six of eight surviving correction); 2–7 of 17 conditions per pair named significantly less often (Table 10)	published must-not-miss sets (+9 to +24 points in seven of eight settings)	worked examples (mean falls for Qwen2.5-32B and OLMo 3 7B); averaged lists; with published sets only OLMo, Mistral and few-shot Qwen2.5 lose specific conditions
(f) Asking for must-not-miss conditions raises coverage, partly by lengthening the list (§4.6)	+6 to +15 points of warranted severe conditions named, 0.01–0.11 beyond two combined standard lists (Table 18)	published must-not-miss sets (+15 to +26 points; 8–20 beyond two lists)	costs one to two more unwarranted severe conditions per list
(g) Voting stops helping and agreement weakens as an error signal (§4.7)	Voting +3.4 to +8.4 points for base models, at most +1.9 for post-trained ones; agreement's AUROC falls in every family (Table 8)	worked examples; Mistral; rare-disease cases	case reports (voting still helps; agreement weakens only for Qwen3-8B); models that reason first; thinking mode; Qwen2.5-7B-Instruct at T=1.3; agreement drop n.s. for OLMo
(h) The better error signal depends on model and prompt; combining helps post-trained models modestly (§4.7)	Held-out audit over 56 model–setting combinations: logistic combination +0.030 $[0.013, 0.044]$ over 31-sample agreement for post-trained models, rank average +0.019 (n.s.) (§4.19)	worked examples (stated probability better for three post-trained models)	base models and thinking mode (agreement alone as good or better); rare-disease cases and case reports (agreement better)
(i) On published case reports the sampled differential narrows	Samples cover 3–14 points fewer of the authors'	strict grading (judges agree on 97.6%, κ	pass@32 loss significant only for Qwen3-8B and

Control	(a) View kept, variation removed	(b) Samples collapse	(c) First-answer gain is sampling noise	(d) Stated list improves (smaller Qwen)	(e, f) Severe conditions; must-not-miss prompt	(g) Voting and agreement weaken
Worked examples, reference lists (§4.10)	holds	holds, larger (15–30 pts)	–	holds for Qwen3-8B, Qwen2.5-7B; TV only for Qwen2.5-32B; weaker for OLMo 3.1 32B	partly: mean falls for Qwen2.5-32B, OLMo 3 7B	holds
Worked examples from another model (§4.11)	–	holds (13–30 pts)	–	holds for Qwen and Mistral, not OLMo	no excess losses (count test)	holds
Matched temperature (§4.11)	–	Qwen3-8B holds; Qwen2.5-7B-Instruct matches at T=1.3	–	holds	–	same split as (b)
Greedy decoding, matched prompts (§4.12)	–	–	holds in 4 of 6 pairs; not Qwen3-8B, nor Mistral with shared examples	holds for Qwen3-8B, Qwen2.5-7B (not Qwen2.5-32B recall); OLMo 3 7B farther	–	–
Temperature frontier (§4.12)	–	holds for Qwen3-8B, OLMo 3 7B; curves meet for Qwen2.5-7B	–	list unchanged across T	–	agreement below base at every T
Must-not-miss prompt (§4.12)	–	–	–	–	coverage +6–15 pts, partly by length	–
Published must-not-miss sets (§4.13)	–	–	–	–	mean up in 7 of 8; specific losses only OLMo, Mistral, few-shot Qwen2.5	–
Format, accuracy, floor, complete lists (§4.9)	–	–	–	holds	–	–
Other families' base models (Table 24)	holds across families; blurs within	–	–	–	–	–
Every other model's view (Table 21)	partly: own base closest for 3 of 5; as close to nearest post-trained model	–	–	–	–	–
Chance-corrected pass@32 (§4.14)	–	holds with examples (7–18 pts); zero-shot 32B loss vanishes	–	–	–	–
Ties, list size, validity, prevalence (§4.15)	–	holds at natural prevalence	–	holds at natural prevalence	–	agreement weaker without ties; top-3 list advantage is list size
32B pairs (§4.5)	holds	n.s. zero-shot; holds with examples	holds (both pairs)	holds for Qwen; partly OLMo	holds for Qwen; OLMo n.s.	partly: agreement drop n.s. for OLMo
Mistral-7B, third family (§4.16)	holds	holds	fails with shared examples	holds, smallest gain	no mean gain; 3 of 17 lost significantly	holds
Thinking mode (§4.17)	holds	fails: pass@32 restored, still narrow	–	holds, closest list	–	fails: agreement better again
Rare-disease cases (§4.23)	–	–	–	–	–	holds; agreement

Pair (post − base)	pass@32 (pts)	Distinct	TV samples	TV one list	TV list w/o gold	DDF1	AUROC agree@32
Qwen3-8B, T=0.7	−12.4	−2.50	+0.043	−0.043	−0.018†	+0.042	−0.160
	[−19.6, −6.0]	[−2.89, −2.13]	[+0.021, +0.064]	[−0.059, −0.028]	[−0.032, −0.005]	[+0.029, +0.056]	[−0.216, −0.096]
Qwen3-8B, T=1	−14.6	−5.06	+0.067	−0.074	−0.039	+0.066	−0.158
	[−21.5, −8.3]	[−5.49, −4.65]	[+0.042, +0.091]	[−0.090, −0.059]	[−0.052, −0.027]	[+0.053, +0.080]	[−0.211, −0.104]
Qwen2.5-7B,	−9.7	−3.09	+0.028†	−0.062	−0.041	+0.062	−0.106
T=0.7	[−16.1, −4.1]	[−3.63, −2.57]	[+0.007, +0.049]	[−0.076, −0.049]	[−0.054, −0.028]	[+0.049, +0.075]	[−0.163, −0.044]
Qwen2.5-7B, T=1	−10.3	−5.59	+0.042†	−0.088	−0.056	+0.083	−0.104
	[−17.2, −4.2]	[−6.03, −5.15]	[+0.015, +0.068]	[−0.103, −0.074]	[−0.069, −0.043]	[+0.069, +0.097]	[−0.159, −0.046]
Qwen2.5-32B,	−4.9†	−2.10	+0.033	−0.053	−0.043	+0.051	−0.079
T=0.7	[−10.3, +0.5]	[−2.49, −1.74]	[+0.018, +0.049]	[−0.064, −0.042]	[−0.053, −0.032]	[+0.041, +0.061]	[−0.119, −0.038]
OLMo 3 7B,	−28.3	−2.56	+0.090	+0.029†	+0.034	−0.006†	−0.076†
T=0.7	[−35.6, −21.1]	[−2.84, −2.27]	[+0.071, +0.109]	[+0.004, +0.052]	[+0.017, +0.051]	[−0.022, +0.009]	[−0.157, +0.001]
OLMo 3 7B, T=1	−15.2	−3.95	+0.079	−0.009†	+0.006†	+0.024	−0.076†
	[−20.8, −9.7]	[−4.22, −3.68]	[+0.060, +0.098]	[−0.031, +0.011]	[−0.006, +0.018]	[+0.010, +0.036]	[−0.151, −0.011]
OLMo 3.1 32B,	−4.0†	−2.18	+0.020†	−0.045	−0.023	+0.040	−0.056†
T=0.7	[−6.8, −1.2]	[−2.47, −1.90]	[+0.002, +0.037]	[−0.059, −0.031]	[−0.034, −0.012]	[+0.026, +0.054]	[−0.095, −0.019]

Table 34: Base → post-trained (final Instruct) differences on DDXPlus with 95% intervals from a bootstrap over the 49 true-pathology clusters. †: not significant after Holm correction across 112 tests (50,000 resamples, add-one p-values). pass@32 here is recomputed from the samples with the analysis scripts' name matching and can differ from Table 2 by up to 0.3 points; the AUROC differences come from the same script and differ from those implied by Table 8 by up to 0.005.

7 Condition-clustered intervals and multiple comparisons

The 112 headline tests below (14 metrics fixed in advance, eight zero-shot settings) are the primary analysis; every control, stage, few-shot and per-condition analysis is secondary, corrected within its own family, and read as a check on them. DDXPlus has 20 test cases for each of 49 true pathologies, and effects can be specific to a condition, so cases are not independent. Table 34 gives the headline base-vs-post differences with a bootstrap that resamples the 49 pathologies with all their cases (50,000 resamples for these headline tests, so that results near the threshold do not flip between runs; 10,000 elsewhere). Two-sided p-values use the add-one correction, $2(1 + \min(n_{\leq 0}, n_{\geq 0}))/(B + 1)$, so the smallest attainable value lies below the strictest Holm threshold, 0.05/112. Holm correction runs over the 112 tests (8 pairs × 14 metrics fixed in advance: pass@1, pass@32, distinct answers, coverage@32, sampled TV, list TV, mass named, severe named, list and sample TV without the true pathology, DDF1, unwarranted severe conditions named, and the two AUROCs; the table shows seven). The same procedure is applied separately to the few-shot pairs (70 tests), the Mistral and thinking-mode pairs (42) and the stage and medical comparisons (154). Not surviving correction among the headline tests shown, marked †: list TV without the true pathology for Qwen3-8B at T=0.7; sampled TV for Qwen2.5-7B at T=0.7; sampled TV for Qwen2.5-7B at T=1; pass@32 for Qwen2.5-32B at T=0.7; list TV for OLMo 3 7B at T=0.7; DDF1 for OLMo 3 7B at T=0.7; agreement AUROC for OLMo 3 7B at T=0.7; list TV for OLMo 3 7B at T=1; list TV without the true pathology for OLMo 3 7B at T=1; agreement AUROC for OLMo 3 7B at T=1; pass@32 for OLMo 3.1 32B at T=0.7; sampled TV for OLMo 3.1 32B at T=0.7; agreement AUROC for OLMo 3.1 32B at T=0.7. OLMo 3 7B's list TV without the true pathology at T=0.7 survives as a significant *increase*.

Pooled effects and equivalence. Statements such as "robust only for Qwen" compare significance across pairs. Table 35 instead pools each difference across pairs with a random-effects model and tests whether Qwen differs from the other families. The sampled collapse, the stated-list gain in recall and the weaker agreement are significant when pooled in all three prompt settings, and no family difference is detectable for pass@32, recall or agreement; agreement's AUROC falls by 0.09 zero-shot and 0.18–0.19 with worked examples. The family difference in the stated list's TV is

Post-trained − base	Zero-shot (5 pairs) pooled [95% CI]	*p*, Qwen vs other	Worked examples (6) pooled [95% CI]	*p*, Qwen vs other	Another model's examples (6) pooled [95% CI]	*p*, Qwen vs other
pass@32 (points)	−11.5 [-19.0, -3.9]	0.558	−19.3 [-23.9, -14.7]	0.496	−17.7 [-22.2, -13.1]	0.457
distinct answers	−2.45 [-2.73, -2.16]	0.602	−3.84 [-4.96, -2.72]	0.369	−3.82 [-5.09, -2.56]	0.047
recall@5, one list (points)	+11.3 [+6.6, +15.9]	0.249	+13.6 [+7.5, +19.7]	0.543	+13.0 [+7.6, +18.4]	0.587
TV, one list	−0.037 [-0.059, -0.015]	0.231	−0.044 [-0.075, -0.013]	0.001	−0.048 [-0.072, -0.025]	0.015
conditional TV	−0.036 [-0.065, -0.007]	0.080	+0.003 [-0.043, +0.050]	0.211	−0.057 [-0.087, -0.028]	0.301
severe named (points)	+7.2 [+4.4, +9.9]	0.023	+5.1 [+1.3, +9.0]	0.006	+4.8 [+0.9, +8.6]	0.067
AUROC, agreement	−0.092 [-0.127, -0.057]	0.087	−0.194 [-0.240, -0.148]	0.139	−0.183 [-0.212, -0.155]	0.961
AUROC, stated p	+0.081 [-0.017, +0.179]	0.248	+0.029 [-0.014, +0.071]	0.003	+0.052 [+0.004, +0.101]	0.319

Table 35: Pooled base → post-trained effects across pairs (DDXPlus, T=0.7): DerSimonian–Laird random-effects meta-analysis of the per-pair differences, with standard errors from their condition-clustered intervals. *p, Qwen vs other*: test that the pooled effect differs between the Qwen pairs and the OLMo and Mistral pairs (Q_{between}, 1 df). With five or six pairs these tests have little power; they replace counts of significant pairs, not a full model of heterogeneity.

significant with worked examples (p=0.001; 0.015 with another model's examples) but not zero-shot (p=0.23), and severe conditions gain more for Qwen (p=0.023 zero-shot, 0.006 with worked examples). For claims of no difference we use two one-sided tests with margins fixed in advance (5 points for accuracy and recall, 0.02 for TV, 0.03 for AUROC). Thinking mode's pass@32 is at least its base model's (90% interval $[+2.6, +14.1]$ points), but the non-significant zero-shot pass@32 losses at 32B and OLMo 3 7B's recall change ($+1.5$, $[-2.9, +5.9]$) are inconclusive rather than equivalent to zero.

8 Prompts, vignettes and grading

DDXPlus. Each patient's coded evidence is rendered by a fixed template into a vignette of the patient's answers and relevant history; we fix known mistranslations in the English evidence descriptions. An example (test case with true pathology chronic rhinosinusitis; the reference differential gives viral pharyngitis 0.17, chronic rhinosinusitis 0.16, cluster headache 0.15, acute rhinosinusitis 0.14, bronchitis 0.14, URTI 0.12; cluster headache is implausible for this patient, a reminder that the reference is an engine's posterior rather than a clinician's differential):

> `A 13-year-old female presents for evaluation. Symptoms (patient answers): Do you have pain somewhere, related to your reason for consulting? Yes. Characterize your pain: burning. Do you feel pain somewhere? mouth, forehead, cheek(R), cheek(L), nose. How intense is the pain? (scale 0-10) 3. […] Have you lost your sense of smell? Yes. Do you have nasal congestion or a clear runny nose? Yes. Do you have a cough? Yes. Relevant history and risk factors: Have you had a cold in the last 2 weeks? Yes. […] Do you have polyps in your nose? Yes. […]`

System prompt: `You are an experienced emergency and primary care physician.` The user turn is the vignette followed by: `Based on the patient's answers above, give your differential diagnosis. Choose ONLY from this list of possible conditions: [the 49 conditions, one per line]. List the 5 most likely conditions from the list, most likely first, with your probability that each is the correct diagnosis. Use the condition names exactly as written in the list. End your answer with a JSON array in exactly this format: [{"diagnosis": "<name>", "probability": <number between 0 and 1>}, ...].` The full-distribution prompt (§4.8) instead asks for a probability on every condition, summing to one, most likely first. The plain and CoT variants are described in

§4.3. Base models receive the same chat-formatted prompt; OLMo base models use their SFT checkpoint's chat template.

Phenopacket Store. The vignette lists the case's present and (up to ten) excluded HPO phenotypes with sex and age. System prompt: `You are an expert clinical geneticist and diagnostician.` Instructions: `List the 5 most likely specific diagnoses (use the specific disease or subtype name, e.g. 'Spastic paraplegia 7' rather than 'hereditary spastic paraplegia', when you can), most likely first, with your probability that each is the correct diagnosis`, followed by the same JSON format.

Grading open-ended names. Each predicted name is mapped to Mondo by exact lookup against Mondo labels and synonyms (also trying any parenthesized abbreviation and the name without it), then by fuzzy matching (token-sort ratio ≥ 92). It is *exact* if it maps to the Mondo class of the case's OMIM diagnosis, and *related* if it maps to a specific ancestor group of it (at most two is-a steps up, with at most 150 descendants) or to one of its subtypes; siblings sharing only a broad parent are wrong. The grader passes hand-written checks (e.g. "Hereditary spastic paraplegia type 7 (SPG7)" is exact for SPG7, "Hereditary spastic paraplegia" is related, and Turner syndrome is wrong for Jacobsen syndrome), but we did not validate it against clinician grading, so errors at the scale of the 1–2 point effects we report on this dataset are possible.

9 Limitations

DDXPlus is synthetic: its reference differential is the posterior of a rule-based engine, and the simulated pathology is not always ranked first in it (it is for "more than 70%" of patients [8, §3.5]), so we treat TV and coverage as distance to a *reference* posterior rather than to clinical truth. Base models more often give invalid answers. We count these as errors, which makes the large-k losses conservative but inflates post-trained gains at small k; conditioning on valid samples removes a quarter to a third of the T=1 pass@1 gain (§4.3). A re-run of the five final pairs with three worked examples in every prompt nearly removes invalid answers and confirms the results except OLMo 3.1 32B's stated-list gain (§4.10); it uses one temperature and one set of example patients, with lists from the reference or from Qwen2.5-32B-Instruct. Sampled hotter, Qwen2.5-7B-Instruct recovers much of its base model's pass@32 but Qwen3-8B does not (§4.11); the temperature control covers two 7–8B models. Our base/post-trained comparisons cover two open families at 7–32B, plus Mistral-7B with few-shot prompts only; prompt robustness of the sampled readout is tested on one pair, and the collapse at large k largely disappears when the model is asked to reason first (§4.3) or uses its thinking mode (§4.17). The OLMo base models give invalid answers on 27–47% of samples, so their format-controlled comparisons are weaker than Qwen's (below 0.5% with few-shot prompts). We evaluate one RL-Zero checkpoint, at 7B, trained with an LM-judge reward on chat prompts rather than a diagnosis reward, and the 32B models only at T=0.7. Severity is DDXPlus's own label; two published must-not-miss lists mapped onto its 49 conditions give the same average result but locate the specific losses in OLMo and Mistral (§4.13). Phenopacket cases are drawn from published literature and may be in pretraining data, which we partly address with era splits. The stated readout renormalizes each list's probabilities over the conditions it names, and the single-call answer is a model's first valid sample. A full-distribution prompt gives the same picture (§4.8), but other elicitations may differ. Which readout is the better error signal is model-dependent; the audit of §4.19 makes it checkable with 100 labeled cases but does not remove the dependence, and we validate it only within DDXPlus. The test set is class-balanced (20 cases per condition), so

accuracies do not reflect real prevalence, and the prompt lists all 49 conditions, so DDXPlus is a closed-set task. We do not evaluate clinical utility with clinicians.

References

[1] Kasra Arabi, Virginia Smith, and Chhavi Yadav. Sample what you say: Aligning language models to sample the distributions they state, 2026.

[2] Junying Chen, Zhenyang Cai, Ke Ji, Xidong Wang, Wanlong Liu, Rongsheng Wang, and Benyou Wang. Towards medical complex reasoning with LLMs through medical verifiable problems. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 14552–14573, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-acl.751. arXiv version titled "HuatuoGPT-o1, Towards Medical Complex Reasoning with LLMs".

[3] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code. 2021.

[4] Zhoujun Cheng, Shibo Hao, Tianyang Liu, Fan Zhou, Yutao Xie, Feng Yao, Yuexin Bian, Yonghao Zhuang, Nilabjo Dey, Yuheng Zha, Yi Gu, Kun Zhou, Yuqi Wang, Yuan Li, Richard Fan, Jianshu She, Chengqian Gao, Abulhair Saparov, Haonan Li, Taylor W. Killian, Mikhail Yurochkin, Zhengzhong Liu, Eric P. Xing, and Zhiting Hu. Revisiting reinforcement learning for LLM reasoning from a cross-domain perspective. In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*, 2025.

[5] Mehul Damani, Isha Puri, Stewart Slocum, Idan Shenfeld, Leshem Choshen, Yoon Kim, and Jacob Andreas. Beyond binary rewards: Training LMs to reason about their uncertainty. In *International Conference on Learning Representations (ICLR)*, 2026.

[6] Daniel Danis, Michael J. Bamshad, Yasemin Bridges, Andrés Caballero-Oteyza, Pilar Cacheiro, Leigh C. Carmody, Leonardo Chimirri, Jessica X. Chong, Ben Coleman, Raymond Dalgleish, Peter J. Freeman, Adam S. L. Graefe, Tudor Groza, Peter Hansen, Julius O. B. Jacobsen, Adam Klocperk, Maaike Kusters, Markus S. Ladewig, Allison J. Marcello, Teresa Mattina, Christopher J. Mungall, Monica C. Munoz-Torres, Justin T. Reese, Filip Rehburg, Bárbara C. S. Reis, Catharina Schuetz, Damian Smedley, Timmy Strauss, Jagadish Chandrabose Sundaramurthi, Sylvia Thun, Kyran Wissink, John F. Wagstaff, David Zocche, Melissa A. Haendel, and Peter N. Robinson. A corpus of GA4GH phenopackets: Case-level phenotyping for genomic diagnostics and discovery. *Human Genetics and Genomics Advances*, 6(1):100371, 2025. doi: 10.1016/j.xhgg.2024.100371.

[7] Arsene Fansi Tchango, Rishab Goel, Julien Martel, Zhi Wen, Gaetan Marceau Caron, and Joumana Ghosn. Towards trustworthy automatic diagnosis systems by emulating doctors' reasoning with deep reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS 2022)*, volume 35, 2022.

[8] Arsene Fansi Tchango, Rishab Goel, Zhi Wen, Julien Martel, and Joumana Ghosn. DDXPlus: A new dataset for automatic medical diagnosis. In *Advances in Neural Information Processing Systems (NeurIPS 2022), Datasets and Benchmarks Track*, 2022.

[9] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024.

[10] Martha Gulati, Phillip D. Levy, Debabrata Mukherjee, Ezra Amsterdam, Deepak L. Bhatt, et al. 2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR guideline for the evaluation and diagnosis of chest pain. *Circulation*, 144(22):e368–e454, 2021. doi: 10.1161/CIR.0000000000001029.

[11] Paul Hager, Friederike Jungmann, Robbie Holland, Kunal Bhagat, Inga Hubrecht, et al. Evaluation and mitigation of the limitations of large language models in clinical decision-making. *Nature Medicine*, 30(9):2613–2622, 2024.

[12] Xiaoke Huang, Juncheng Wu, Hui Liu, Xianfeng Tang, and Yuyin Zhou. m1: Unleash the potential of test-time scaling for medical reasoning with large language models. In *Proceedings of the Fifth Machine Learning for Health Symposium (ML4H 2025)*, volume 297 of *Proceedings of Machine Learning Research*, pages 369–383. PMLR, 2026.

[13] Chaemin Jang, Dongman Lee, and Jihee Kim. Instruction-tuned language models cannot sample from distributions they can describe. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, 2026.

[14] Ahmadreza Jeddi, Kimia Shaban, Negin Baghbanzadeh, Natasha Sharan, Abhishek Moturu, Elham Dolatabadi, and Babak Taati. When does RL help medical VLMs? disentangling vision, SFT, and RL gains. 2026.

[15] Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. Artificial hivemind: The open-ended homogeneity of language models (and beyond). In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*, 2025.

[16] Zahir Kanjee, Byron Crowe, and Adam Rodman. Accuracy of a generative artificial intelligence model in a complex diagnostic challenge. *JAMA*, 330(1):78, 2023. doi: 10.1001/jama.2023.8288.

[17] Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations (ICLR)*, 2024.

[18] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations (ICLR)*, 2023.

[19] Thom Lake, Eunsol Choi, and Greg Durrett. From distributional to overton pluralism: Investigating large language model alignment. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6794–6814, 2025.

[20] Jixuan Leng, Chengsong Huang, Banghua Zhu, and Jiaxin Huang. Taming overconfidence in LLMs: Reward calibration in RLHF. In *International Conference on Learning Representations (ICLR)*, 2025.

[21] Seungseop Lim, Gibaeg Kim, Hyunkyung Lee, Wooseok Han, Jean Seo, Jaehyo Yoo, and Eunho Yang. H-DDx: A hierarchical evaluation framework for differential diagnosis. 2025. NeurIPS 2025 Workshop on Generative AI for Health (GenAI4Health).

[22] Che Liu, Haozhe Wang, Jiazhen Pan, Zhongwei Wan, Yong Dai, Fangzhen Lin, Wenjia Bai, Daniel Rueckert, and Rossella Arcucci. Beyond distillation: Pushing the limits of medical LLM reasoning with minimalist rule-based RL. 2025. Introduces the AlphaMed model; NeurIPS 2025 Workshop on Generative AI for Health (GenAI4Health).

[23] Mingyi Liu. The alignment tax: Response homogenization in aligned LLMs and its implications for uncertainty estimation, 2026.

[24] Qing Lyu, Kumar Shridhar, Chaitanya Malaviya, Li Zhang, Yanai Elazar, Niket Tandon, Marianna Apidianaki, Mrinmaya Sachan, and Chris Callison-Burch. Calibrating large language models with sample consistency, 2024.

[25] Daniel McDuff, Mike Schaekermann, Tao Tu, Anil Palepu, Amy Wang, Jake Garrison, Karan Singhal, Yash Sharma, Shekoofeh Azizi, Kavita Kulkarni, et al. Towards accurate differential diagnosis with large language models. *Nature*, 642(8067):451–457, 2025. doi: 10.1038/s41586-025-08869-4.

[26] Zhiting Mei, Christina Zhang, Tenny Yin, Justin Lidard, Ola Shorinwa, and Anirudha Majumdar. Reasoning about uncertainty: Do reasoning models know when they don't know? In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 3408–3458, Rabat, Morocco, 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.findings-eacl.178.

[27] Nicole Meister, Carlos Guestrin, and Tatsunori Hashimoto. Benchmarking distributional alignment of large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 24–49, Albuquerque, New Mexico, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.naacl-long.2.

[28] Sonia Krishna Murthy, Tomer Ullman, and Jennifer Hu. One fish, two fish, but not the whole sea: Alignment reduces language models' conceptual diversity. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 11241–11258, 2025. doi: 10.18653/v1/2025.naacl-long.561.

[29] David E. Newman-Toker, Najlla Nassery, Adam C. Schaffer, Chihwen Winnie Yu-Moe, Gwendolyn D. Clemens, et al. Burden of serious harms from diagnostic error in the USA. *BMJ Quality & Safety*, 33(2):109–120, 2024. doi: 10.1136/bmjqs-2021-014130.

[30] Changdae Oh, Qi Zeng, Qi Qi, Andrey Zhmoginov, Deren Lei, Yun He, Hoang Phan, Hangoo Kang, Azalia Mirhoseini, and Sharon Li. Sharpening tax in post-training. 2026.

[31] OpenAI. GPT-4 technical report, 2023.

[32] Isha Puri, Mehul Damani, Idan Shenfeld, Marzyeh Ghassemi, Jacob Andreas, and Yoon Kim. Escaping the mode: Multi-answer reinforcement learning in LMs. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 100127–100146. PMLR, 2026. arXiv version titled "Reaching Beyond the Mode: RL for Distributional Reasoning in Language Models".

[33] Zhiyao Ren, Yibing Zhan, Siyuan Liang, Guozheng Ma, Baosheng Yu, and Dacheng Tao. Towards reliable medical LLMs: Benchmarking and enhancing confidence estimation of large language models in medical consultation, 2026.

[34] Daniel Philip Rose, Chia-Chien Hung, Marco Lepri, Israa Alqassem, Kiril Gashteovski, and Carolin Lawrence. MEDDxAgent: A unified modular agent framework for explainable automatic differential diagnosis. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13803–13826, 2025. doi: 10.18653/v1/2025.acl-long.677.

[35] Thomas Savage, John Wang, Robert Gallo, Abdessalem Boukil, Vishwesh Patel, Seyed Amir Ahmad Safavi-Naini, Ali Soroush, and Jonathan H. Chen. Large language model uncertainty proxies: discrimination and calibration for medical diagnosis and treatment. *Journal of the American Medical Informatics Association*, 32(1):139–149, 2025. doi: 10.1093/jamia/ocae254.

[36] Yuda Song, Julia Kempe, and Remi Munos. Outcome-based exploration for LLM reasoning, 2025.

[37] Jacob Mitchell Springer, Madhu Advani, Lukas Aichberger, Arwen Bradley, Eran Malach, Omid Saremi, Sinead Williamson, Preetum Nakkiran, Etai Littwin, and Aditi Raghunathan. Annotations mitigate post-training mode collapse. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026.

[38] Team Olmo, Allyson Ettinger, Amanda Bertsch, Bailey Kuehl, David Graham, David Heineman, Dirk Groeneveld, Faeze Brahman, Finbarr Timbers, Hamish Ivison, et al. Olmo 3, 2025.

[39] Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

[40] Tao Tu, Mike Schaekermann, Anil Palepu, Khaled Saab, Jan Freyberg, et al. Towards conversational diagnostic artificial intelligence. *Nature*, 642(8067):442–450, 2025.

[41] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

[42] Peter West and Christopher Potts. Base models beat aligned models at randomness and creativity. In *Conference on Language Modeling (COLM)*, 2025.

[43] Fang Wu, Weihao Xuan, Danny Xing, Ximing Lu, Mingjie Liu, Yi Dong, Zaid Harchaoui, and Yejin Choi. The invisible leash: Why RLVR may or may not escape its origin. 2025.

[44] Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. In *International Conference on Learning Representations (ICLR)*, 2024.

[45] Chenghao Yang, Sida Li, and Ari Holtzman. LLM probability concentration: How alignment shrinks the generative horizon. *Transactions on Machine Learning Research*, 2026. arXiv:2506.17871.

[46] Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? In *Advances in Neural Information Processing Systems (NeurIPS 2025)*, 2025.

[47] Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyan Shi. Verbalized sampling: How to mitigate mode collapse and unlock LLM diversity. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 160073–160152. PMLR, 2026.

[48] Chiwei Zhu, Benfeng Xu, Quan Wang, Yongdong Zhang, and Zhendong Mao. On the calibration of large language models and alignment. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. arXiv:2311.13240.